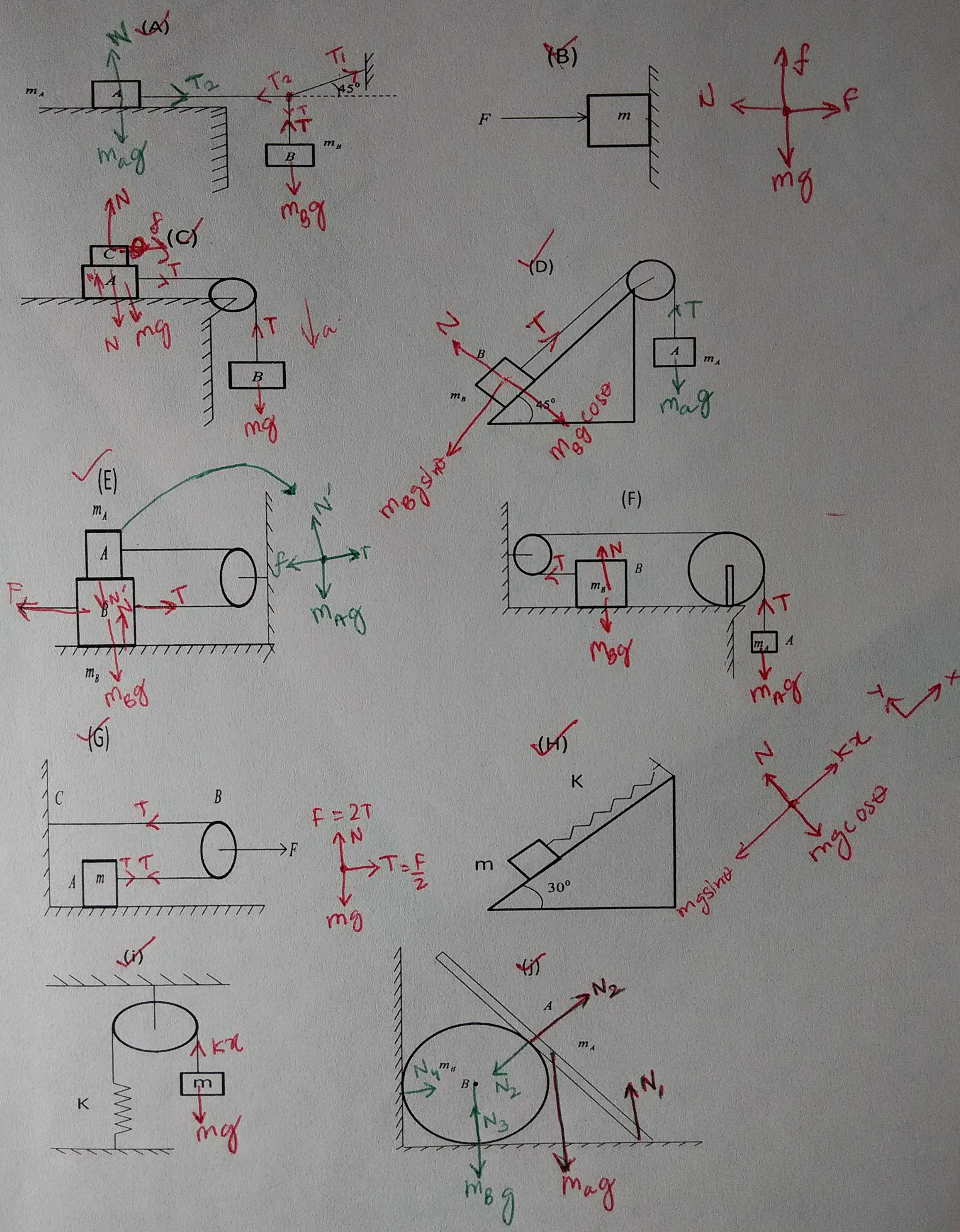
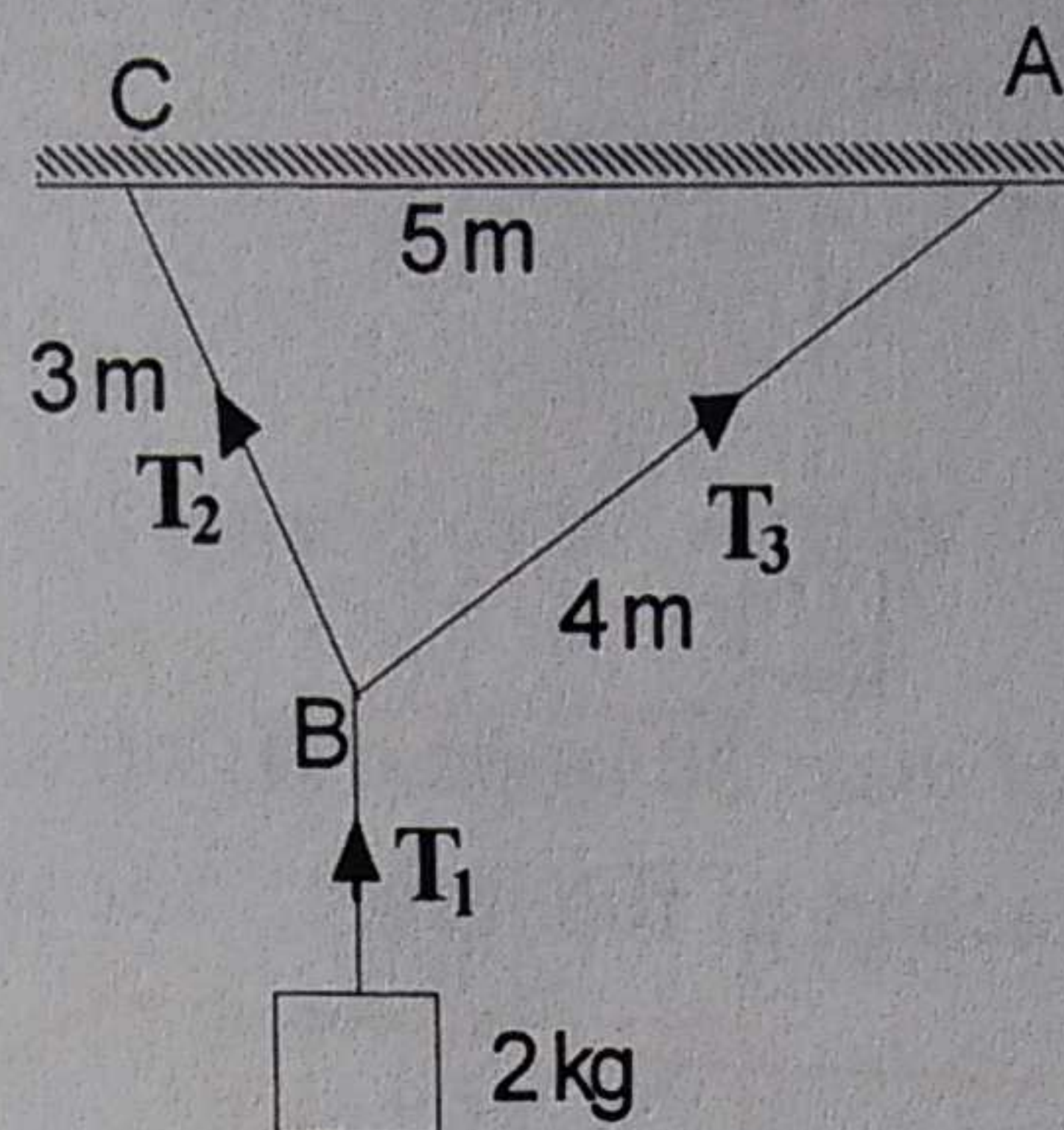


Draw free body diagram of all blocks shown in the systems.

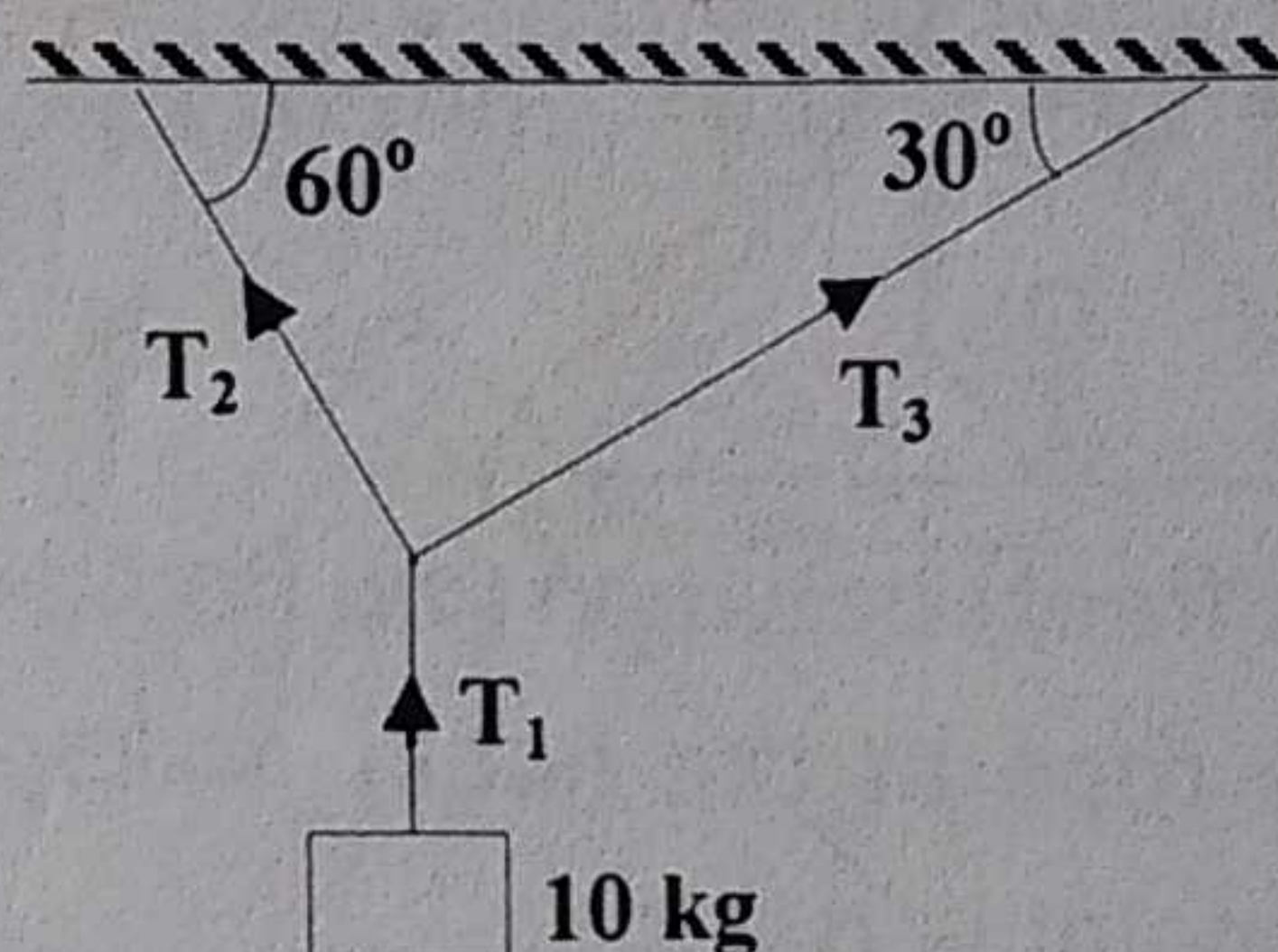


C.P.P OF EQUILIBRIUM OF FORCES

1. For the 2 kg block is equilibrium, the tension in string AB is ($g = 10 \text{ m/s}^2$)

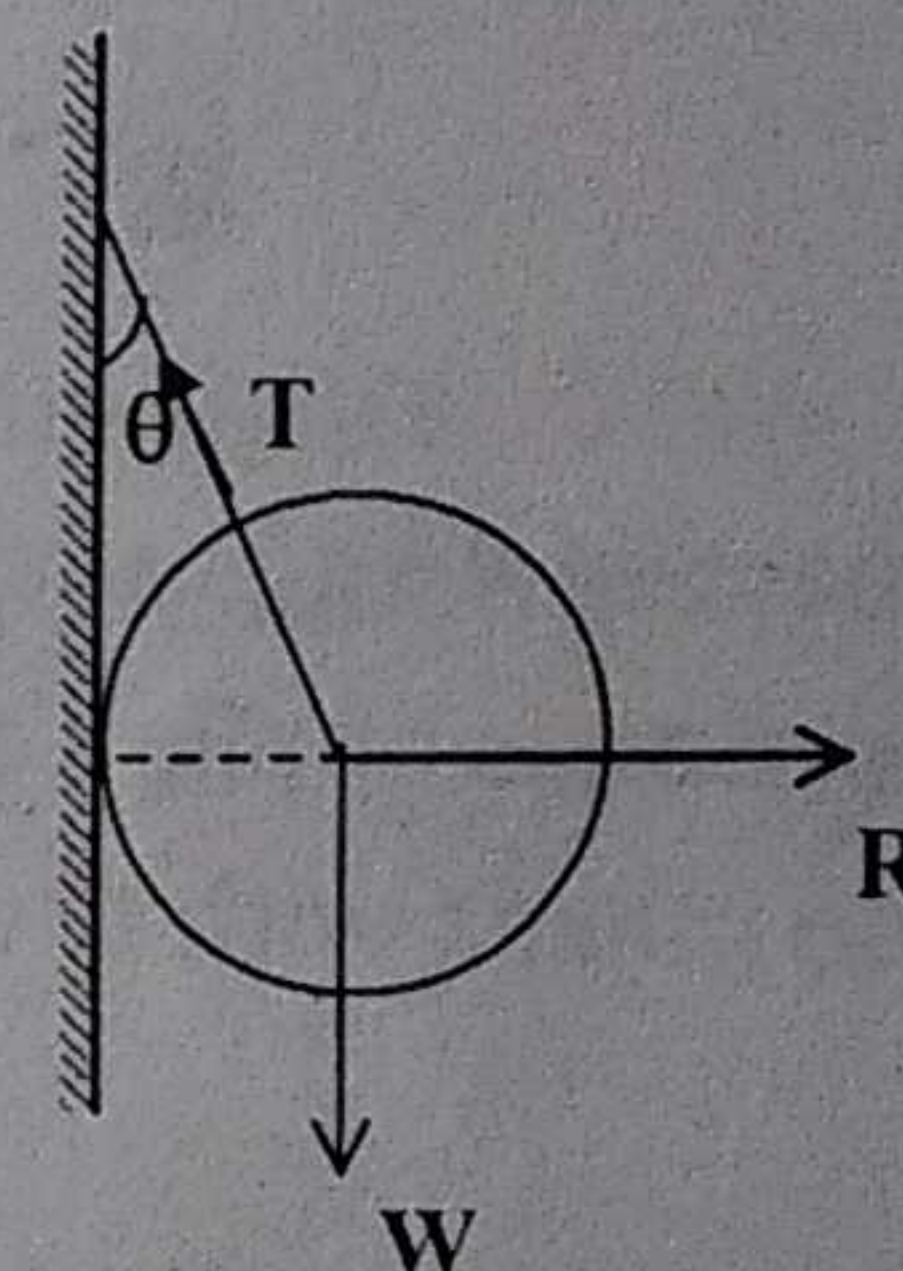


2. A block of mass 10 kg is suspended by three strings as shown in the figure. The tension T_2 is ($g = 10 \text{ m/s}^2$)

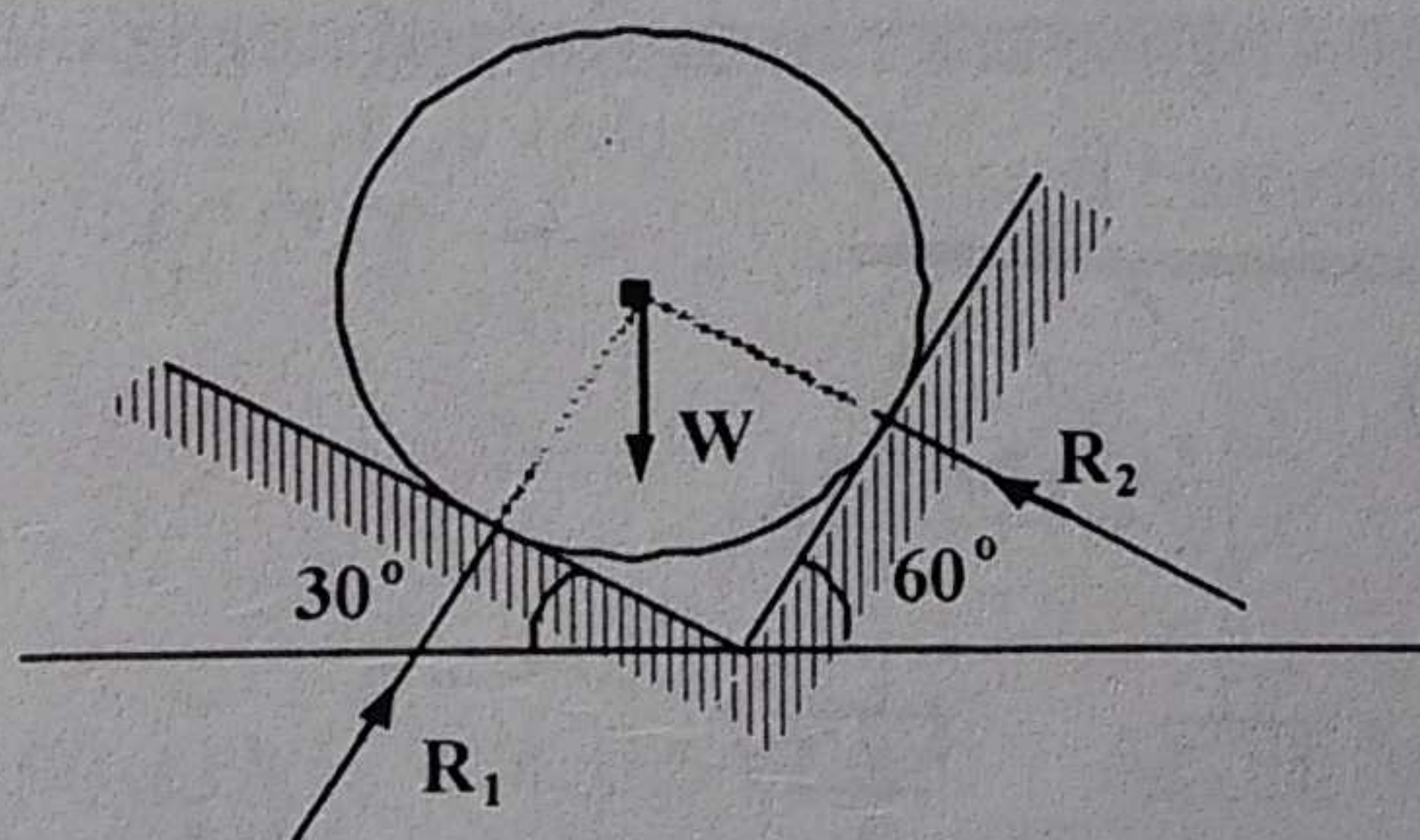


3. A metal of sphere is hung by by a string fixed to a wall. The force acting on the sphere are shown in figure.

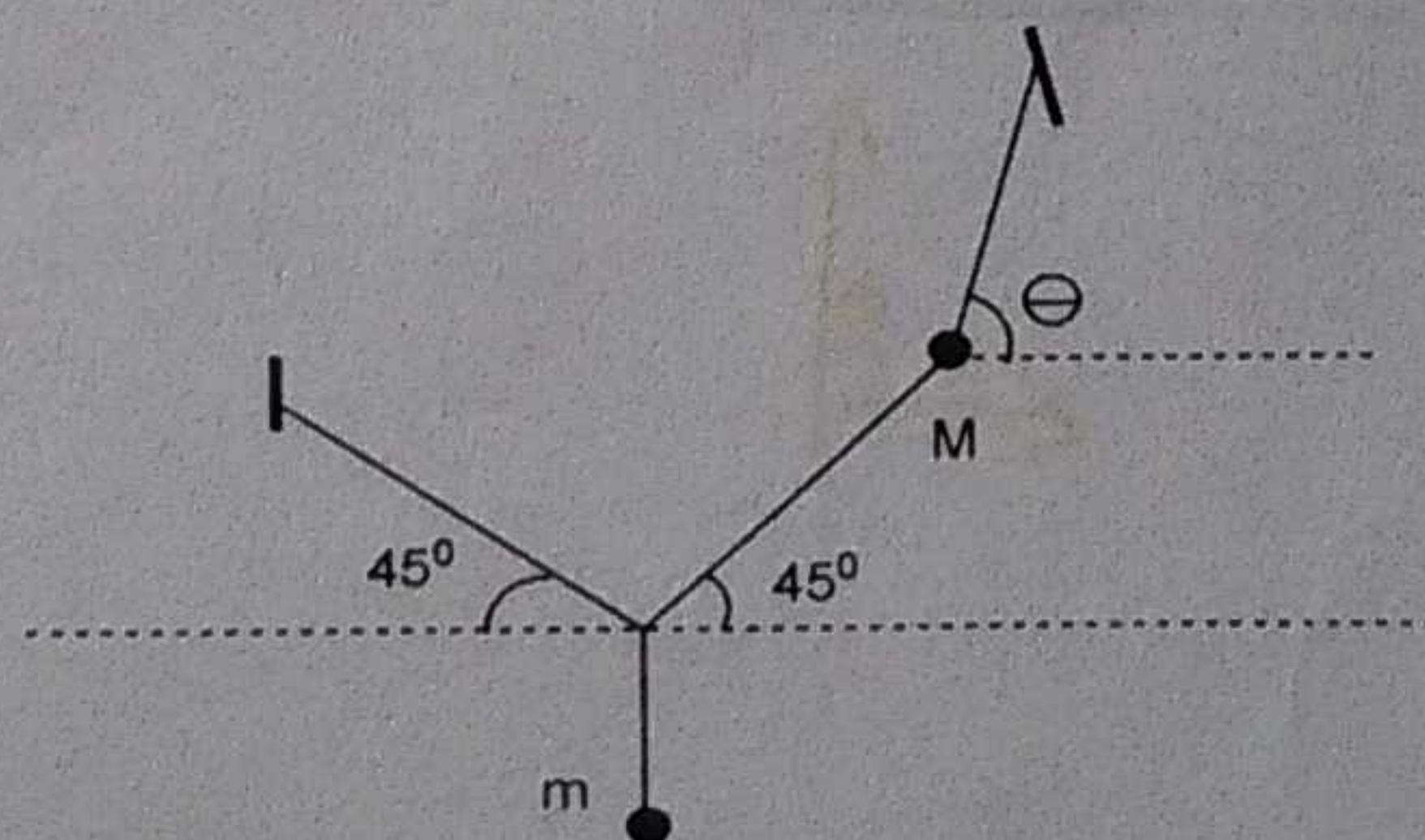
Prove that $T^2 = R^2 + W^2$ and $R = W \tan \theta$.



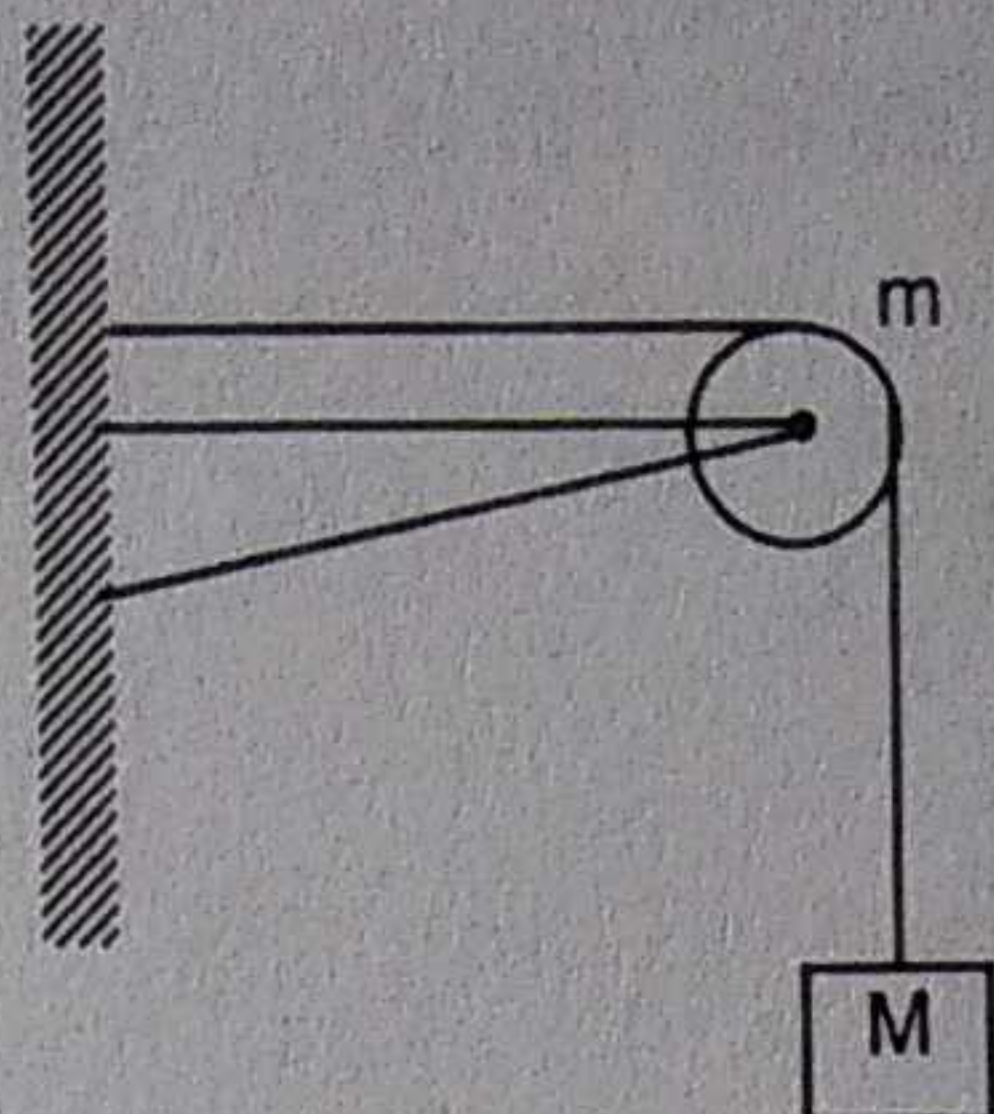
4. A ball of weight W rests on two planes as shown in the figure. What is the magnitude of reaction R_2 ?



5. Two masses m and M are attached with strings as shown. For the system to be in equilibrium prove that $\tan \theta = 1 + \frac{2M}{m}$.



6. A string of negligible mass going over a clamped pulley of mass m supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given by



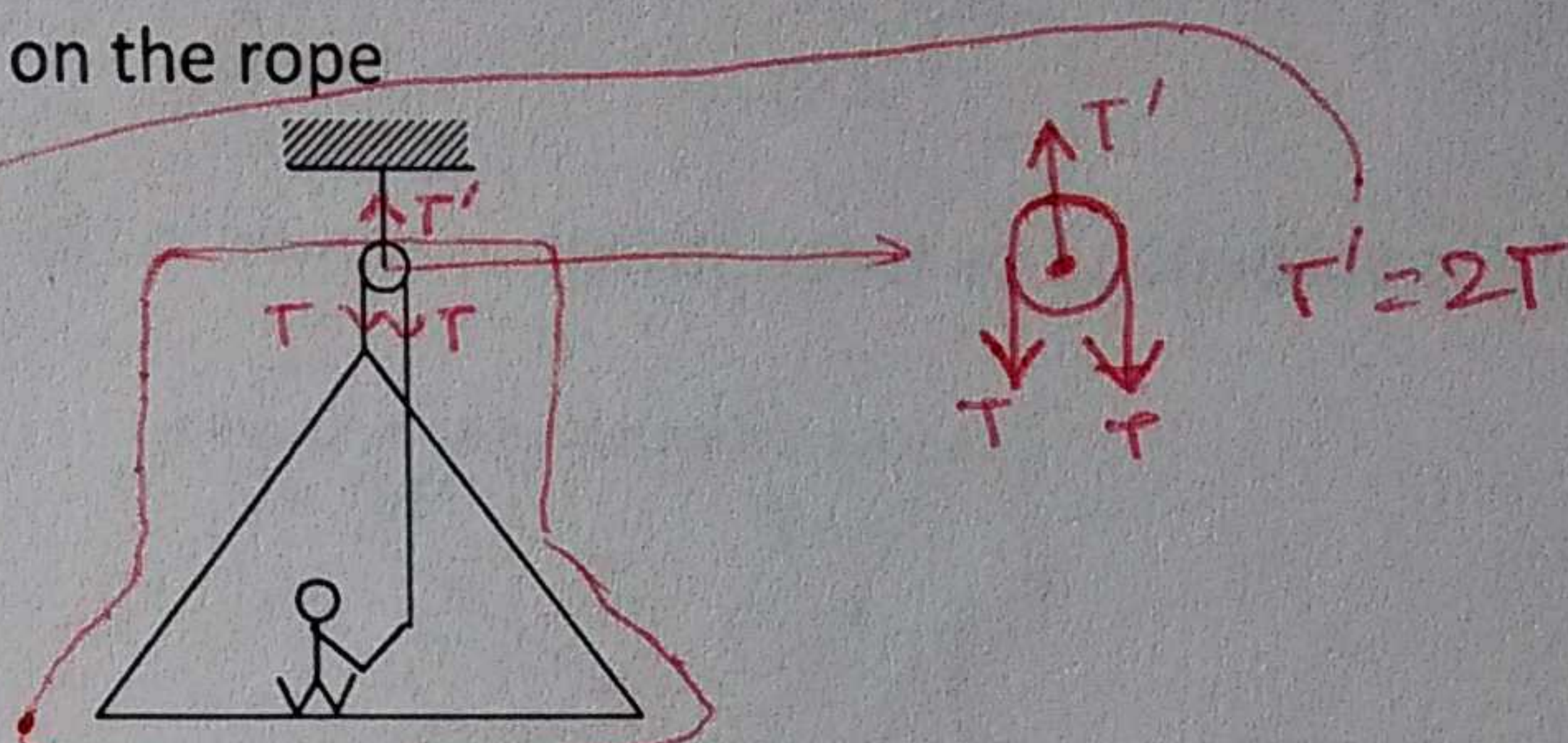
7. A man of mass 50 kg stands on a frame of mass 30 kg he pulls on a light rope which passes over a pulley. The other end of the rope is attached to the frame. For the system to be in equilibrium what force man must exert on the rope

we also know,

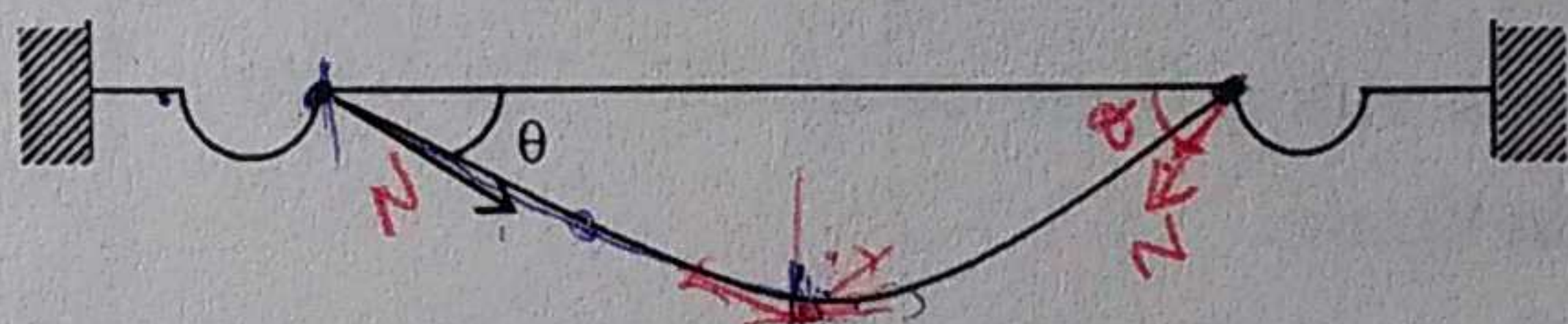
$$T' = (M+m)g$$

$$\Rightarrow \frac{(M+m)g}{2} = T$$

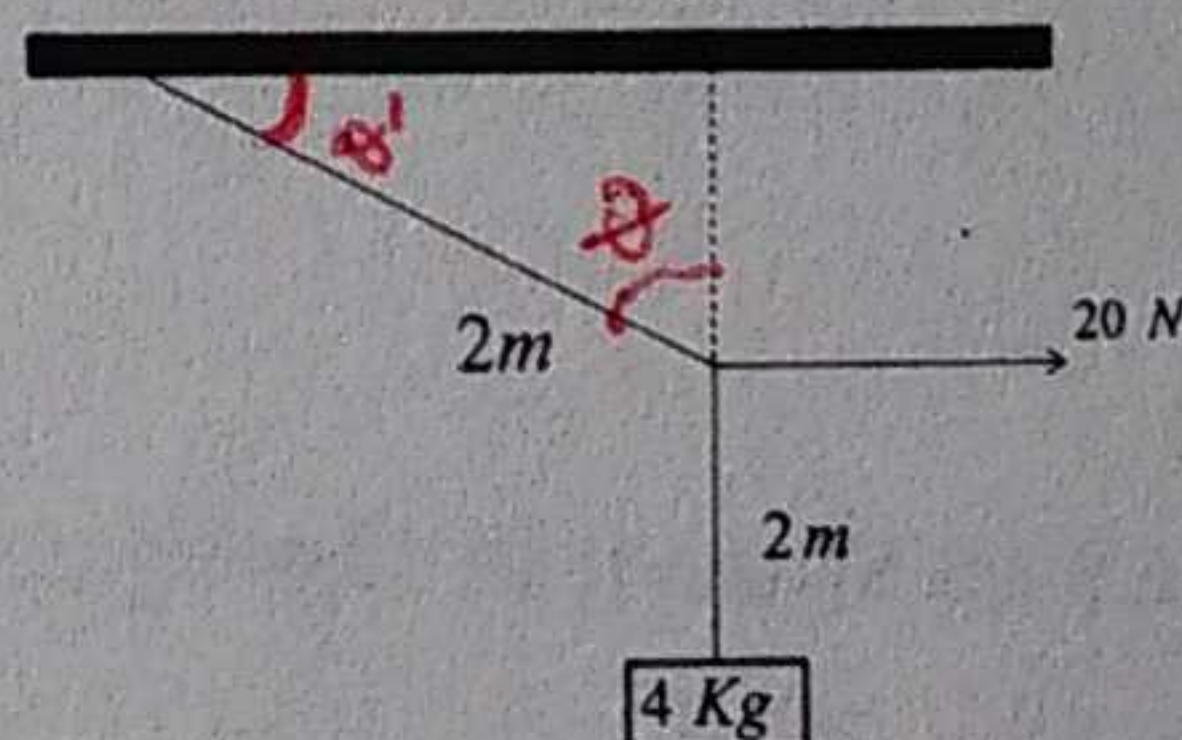
$$\Rightarrow \frac{80g}{2} = 40g$$



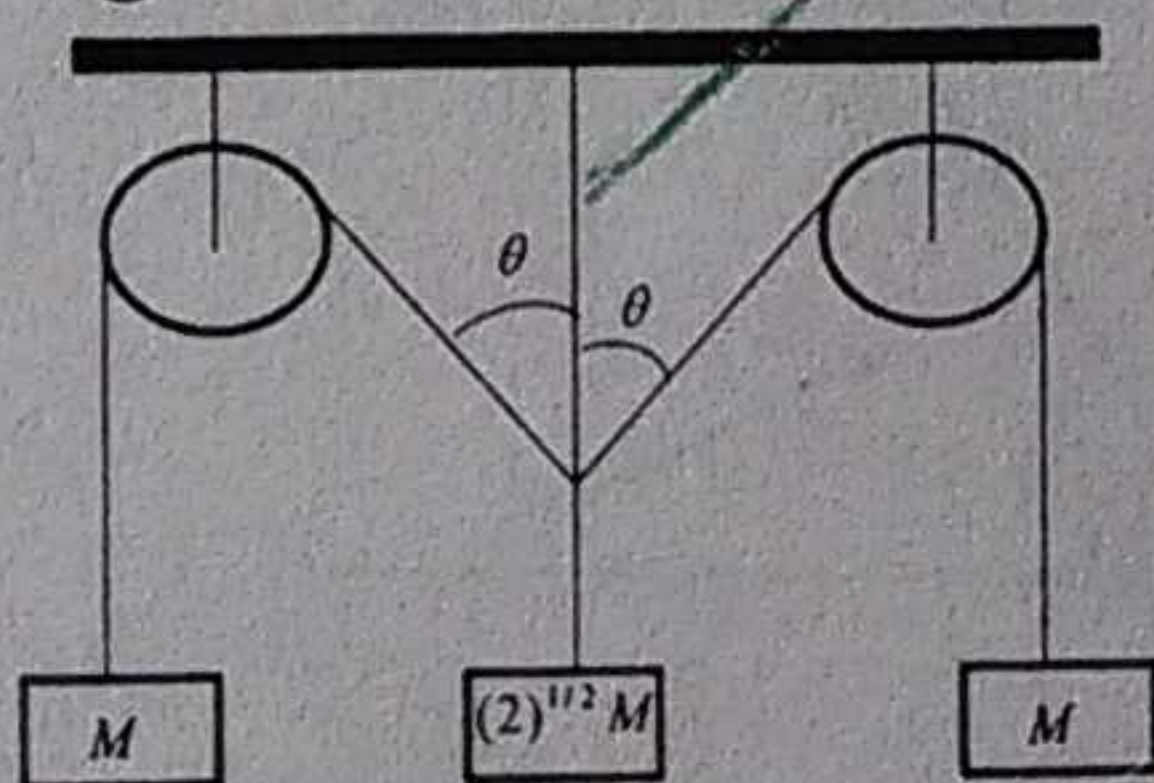
8. A uniform rope of mass m hangs freely between two hooks in the same horizontal line as shown in the figure. The tangent to the rope at the hooks makes an angle θ with the horizontal. The tension in the rope at its lowest point is



9. A mass of 4 kg is suspended by a rope of length 4m from a ceiling. A force of 20 N in the horizontal direction is applied at the mid-point of the rope as shown in figure. What is the angle which the rope makes with the vertical in equilibrium? Neglect the mass of the rope.

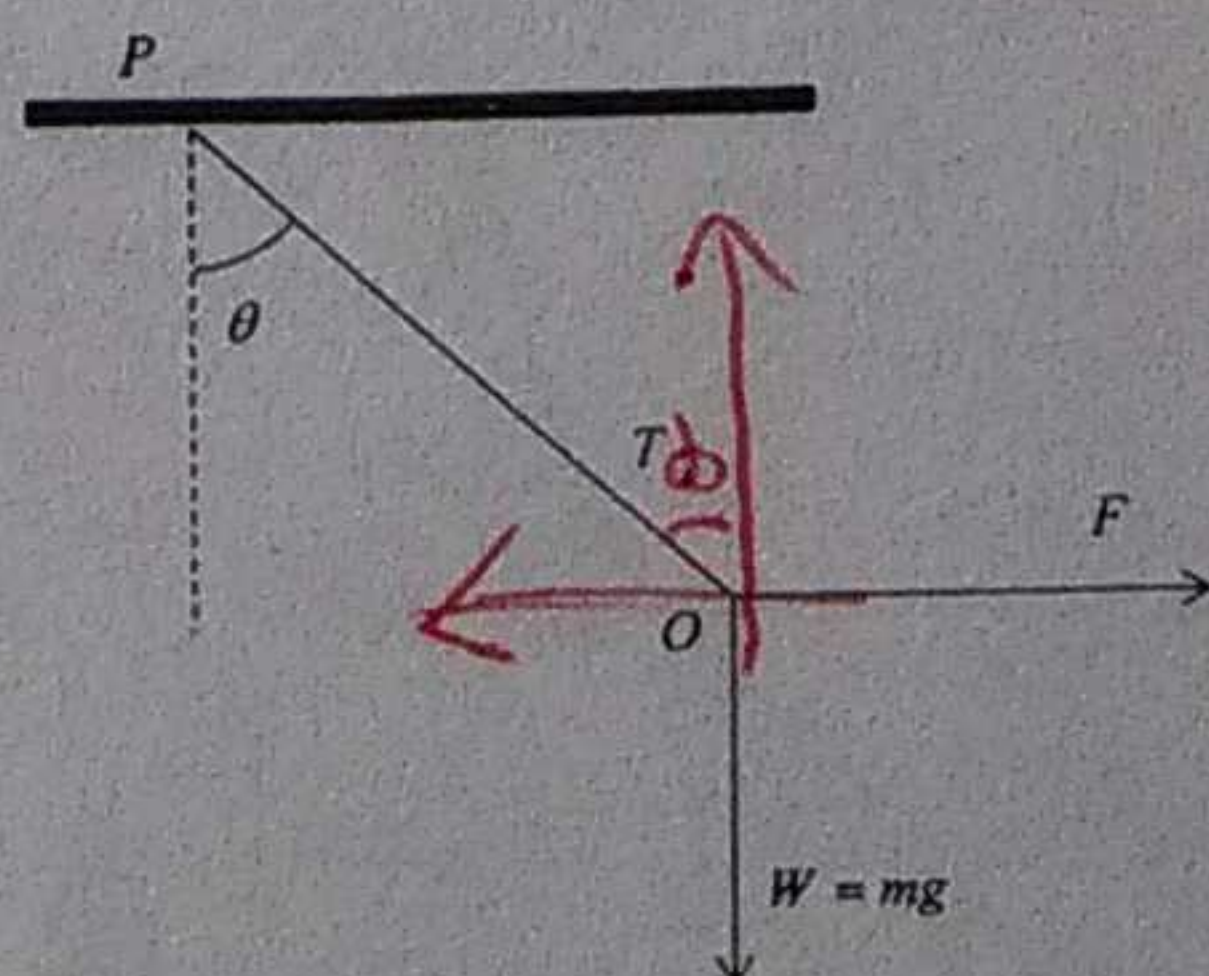


10. The pulleys and strings shown in the figure are smooth and of negligible mass. For the system to remain in equilibrium the angle θ should be:

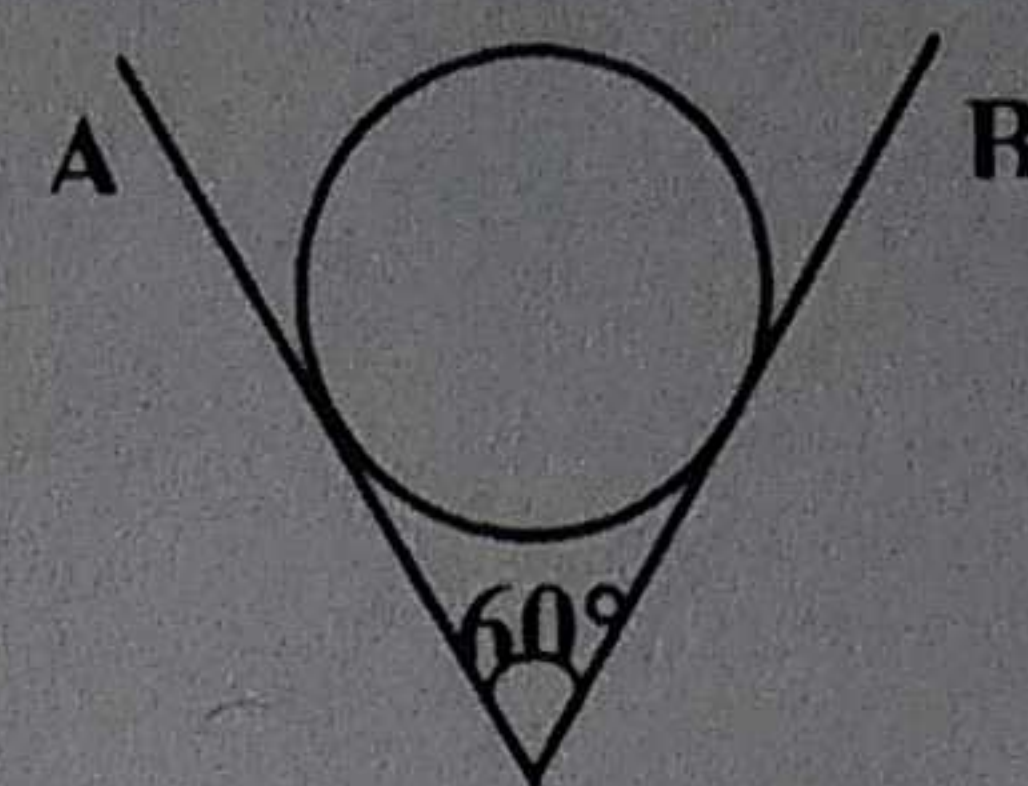


is this a line or string?

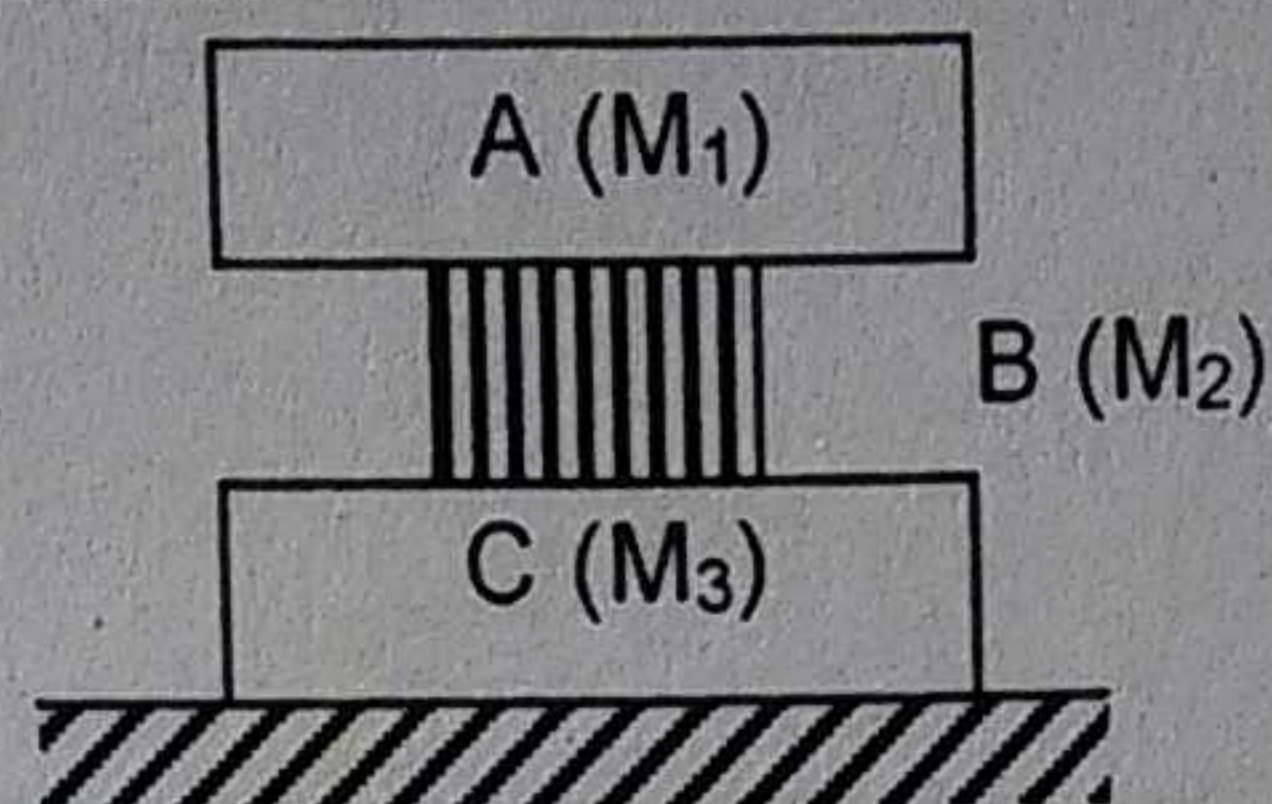
11. A mass m is suspended from a rigid support P by means of a massless string as shown in fig. A horizontal force F is applied at point O of the rope. The system is in equilibrium when the string makes an angle θ with the vertical. If the tension in the string is T and weight is W , then prove that $F = mg \tan \theta$.



12. An iron sphere weights 10 N and rests in a V shaped trough whose sides form a angle 60° . The net force exerted by the walls on the sphere in case as shown in figure is

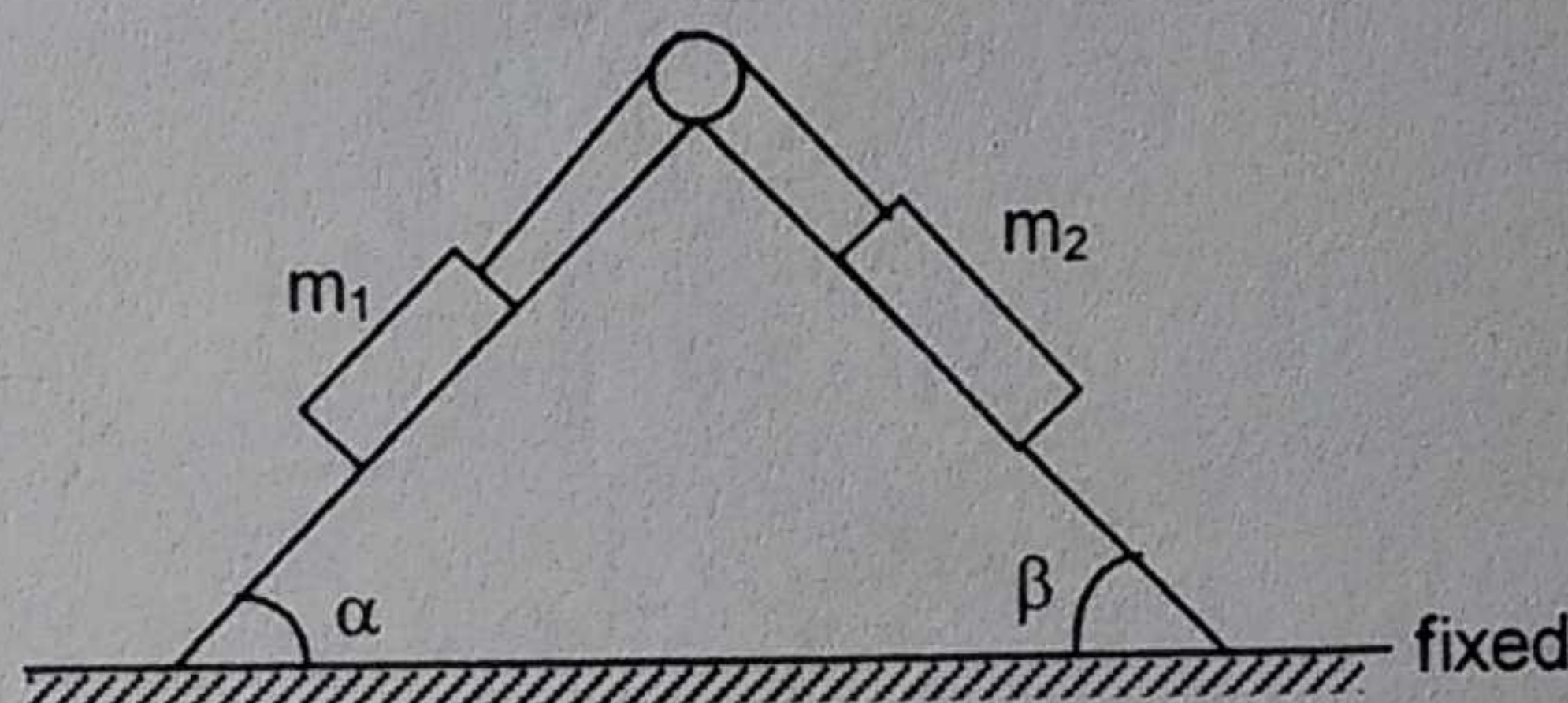


13. Three blocks A, B and C are placed one over the other as shown in figure. Draw the free body diagram of all the three blocks.

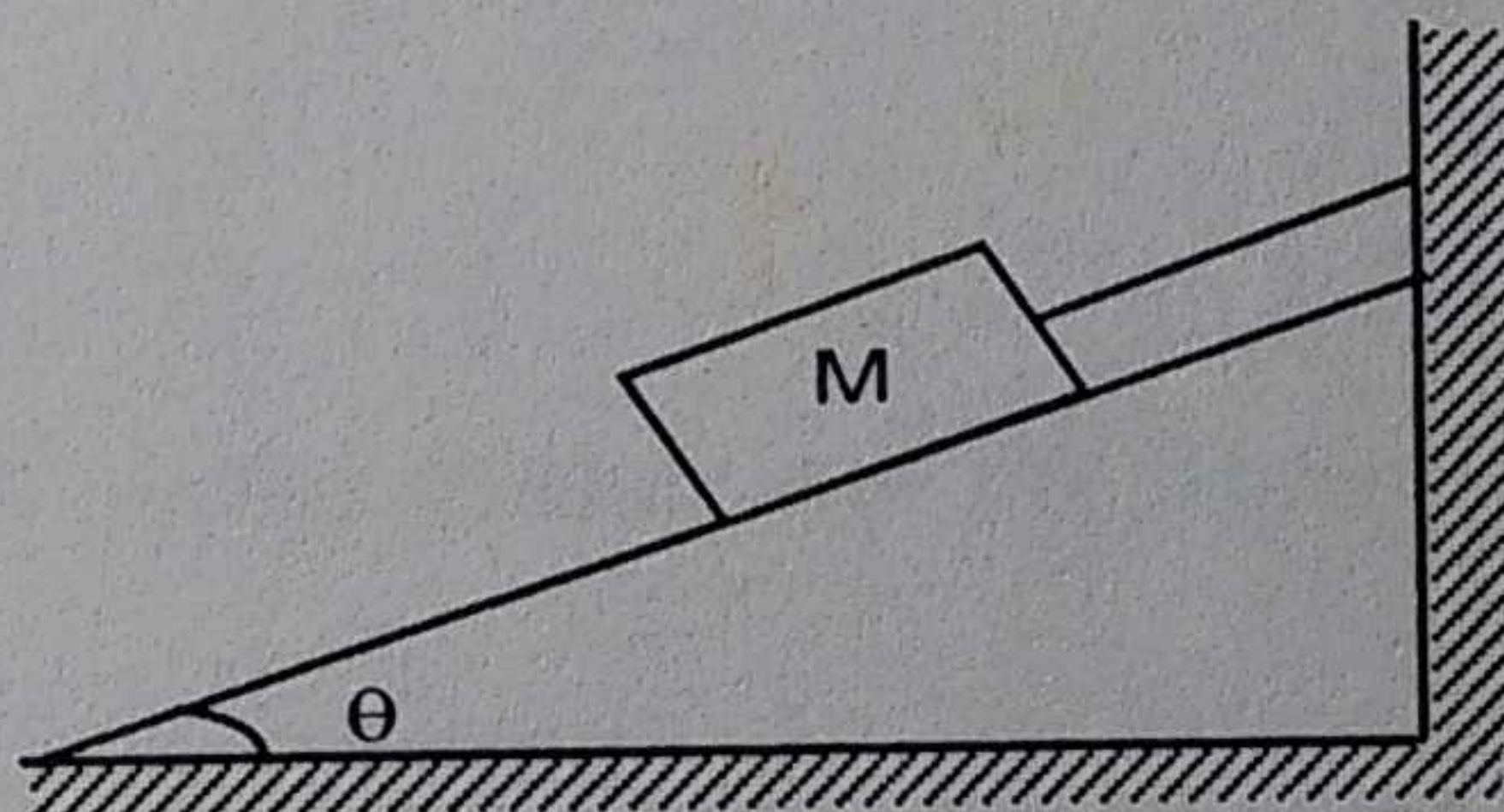


14. (a) Draw the FBD of m_1 and m_2 .

(b) What will the relation between m_1 , m_2 , α , β ? so that m_1 , m_2 remains stationary. All surfaces are smooth.



15. A block of mass M is placed on a smooth inclined surface as shown in the figure. Draw the free body diagram of the mass M . Find tension in the string and the normal reaction exerted by the inclined surface on the block.



ANSWERS

1. 12 N

2. $50\sqrt{3}N$

4. 0.5 W

6. $\sqrt{[(M+m)^2 + M^2]}g = \frac{F}{\sin\theta}$ 7. 40g

8. $\frac{mg}{2\tan\theta}$

9. $\tan^{-1} \frac{1}{2}$

10. 45°

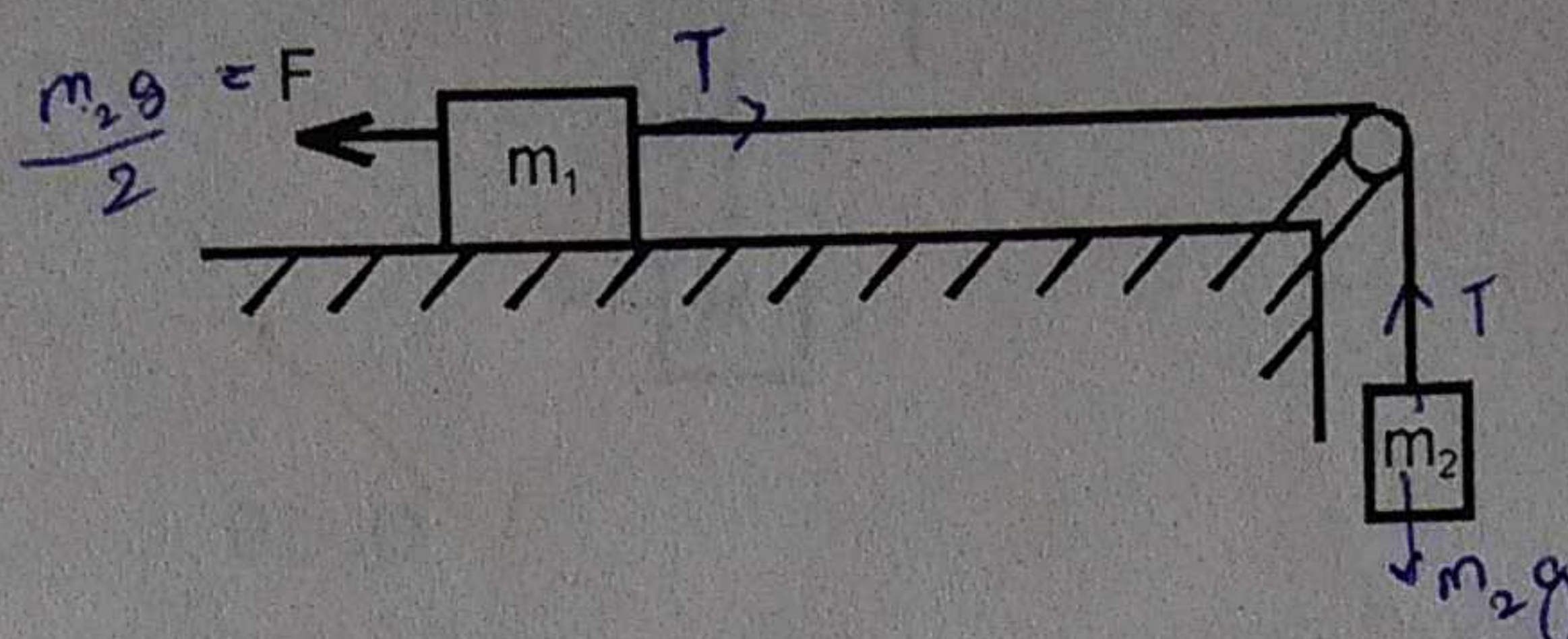
12. 10N

14. $m_1 \sin\alpha = m_2 \sin\beta$

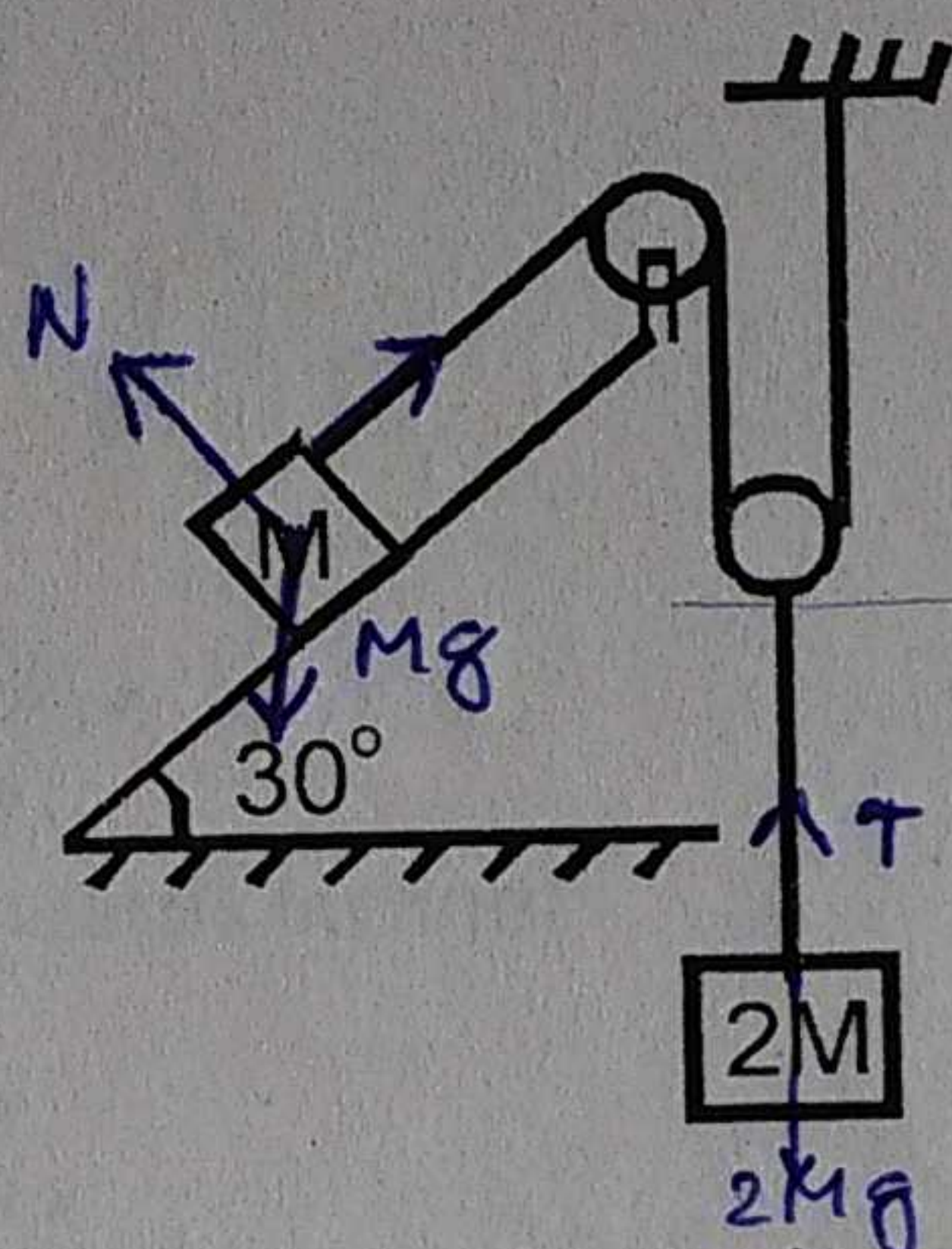
15. $Mg \sin\theta$, $Mg \cos\theta$

LAWS OF MOTION (C.P.P - 1)

- Q-1** A constant force $F = m_2g/2$ is applied on the block of mass m_1 as shown. The string and the pulley are light and the surface of the table is smooth. Find the acceleration of m_1 .

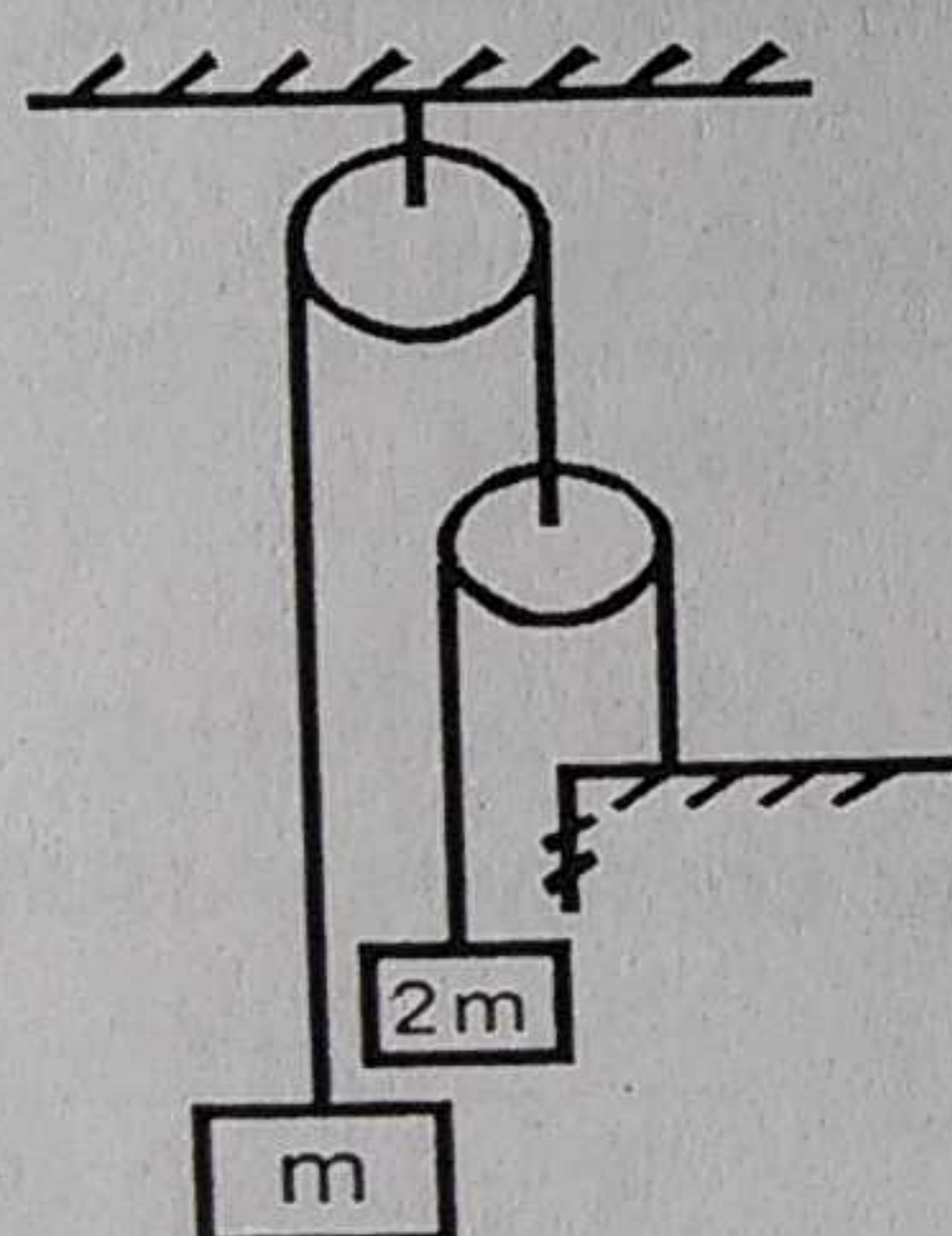


- Q-2** Find the acceleration of the block of mass M in the situation shown in figure. All the surface are frictionless and the pulleys and the string are light.

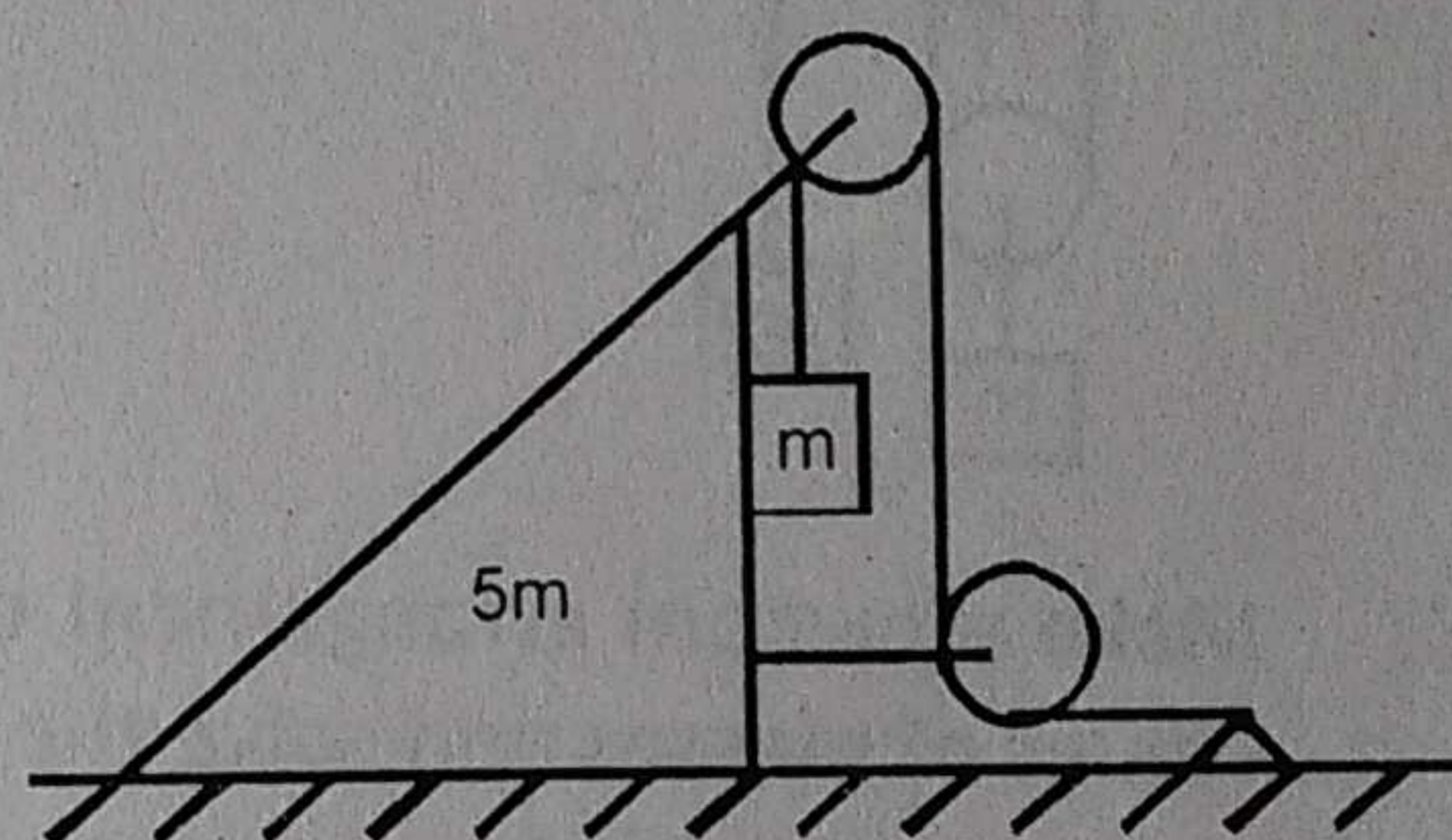


Assuming ideal strings and pulleys and also frictionless surface calculate acceleration of all the masses of the system.

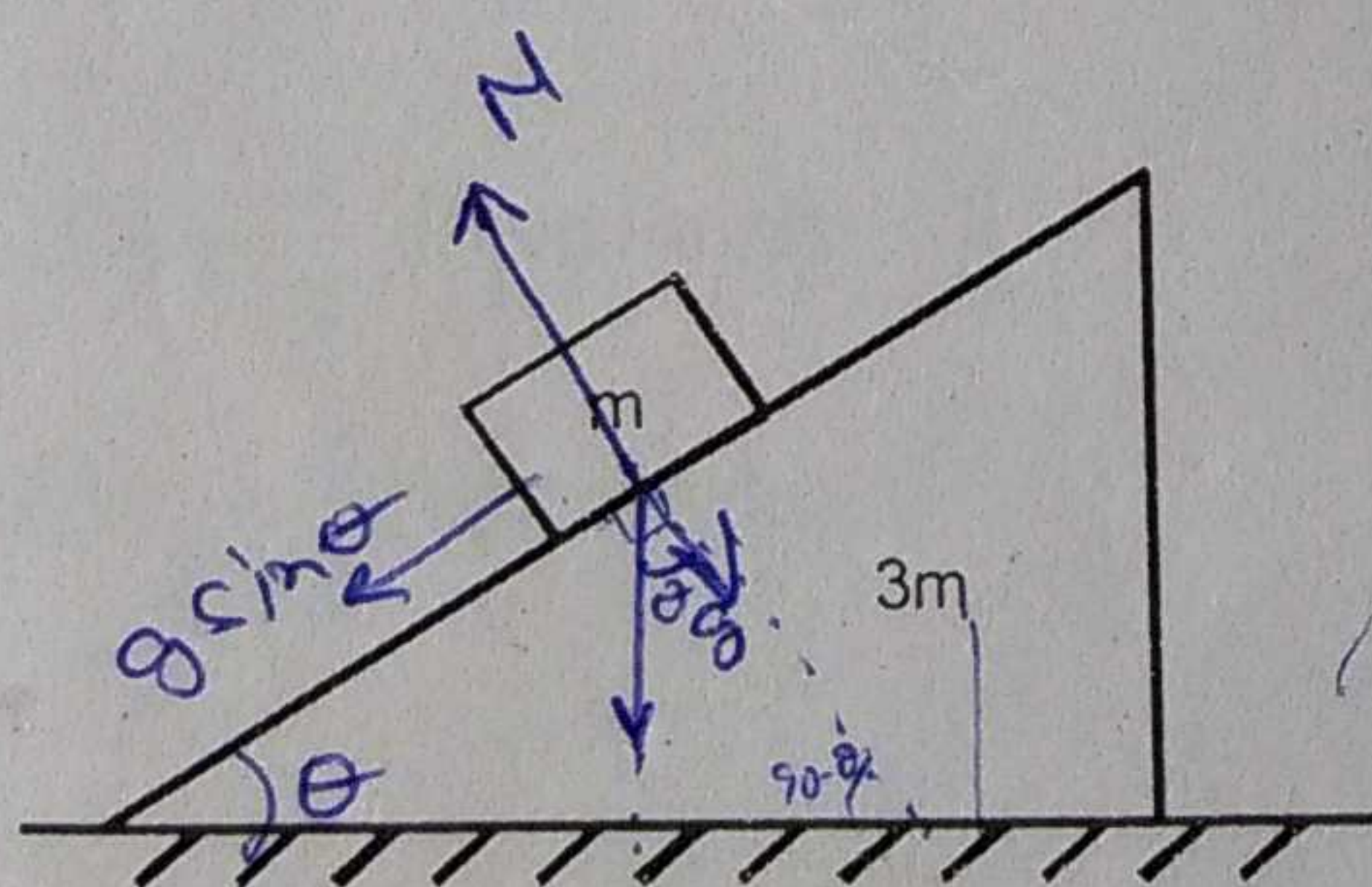
Q-3



Q-4

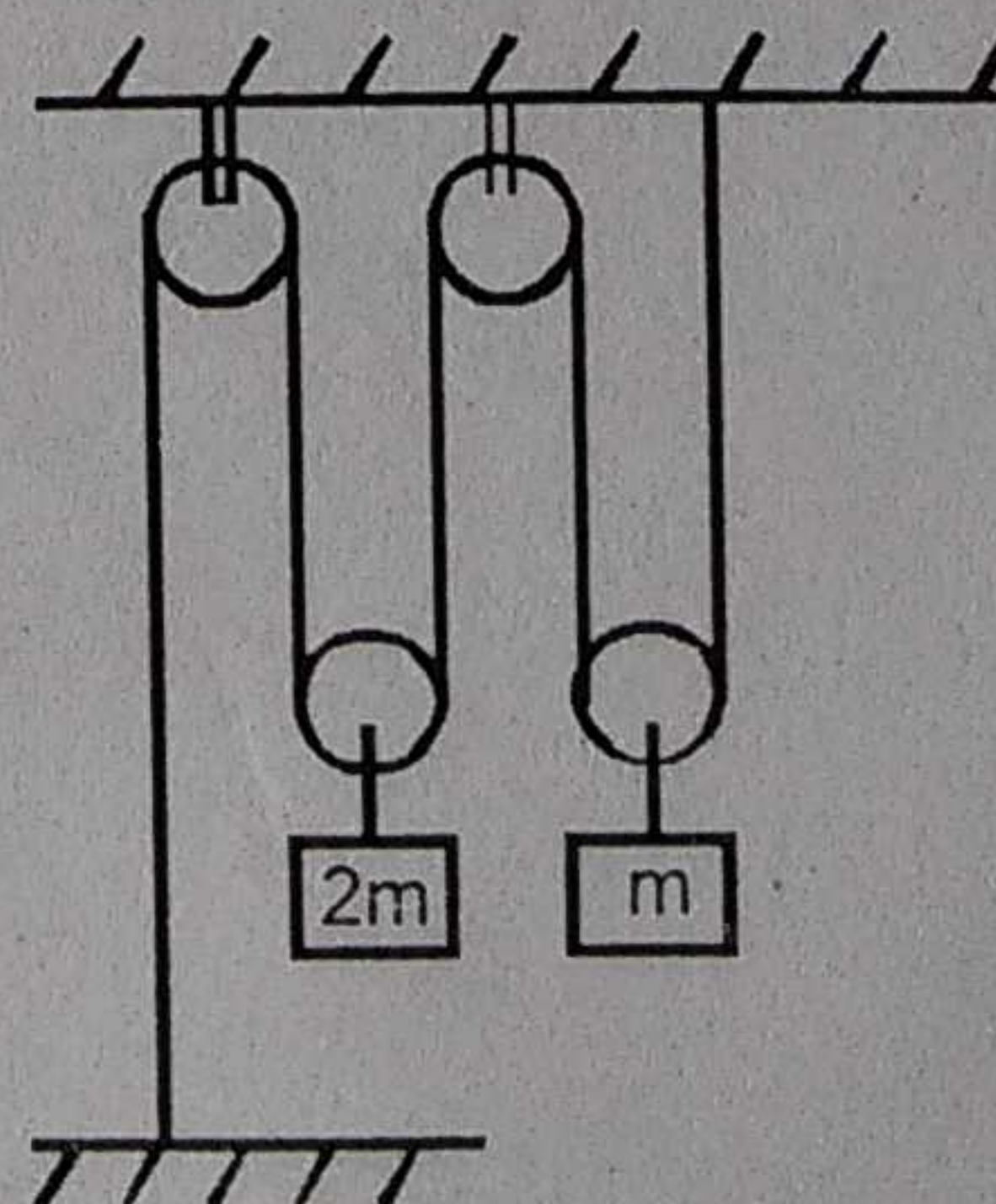


Q-5

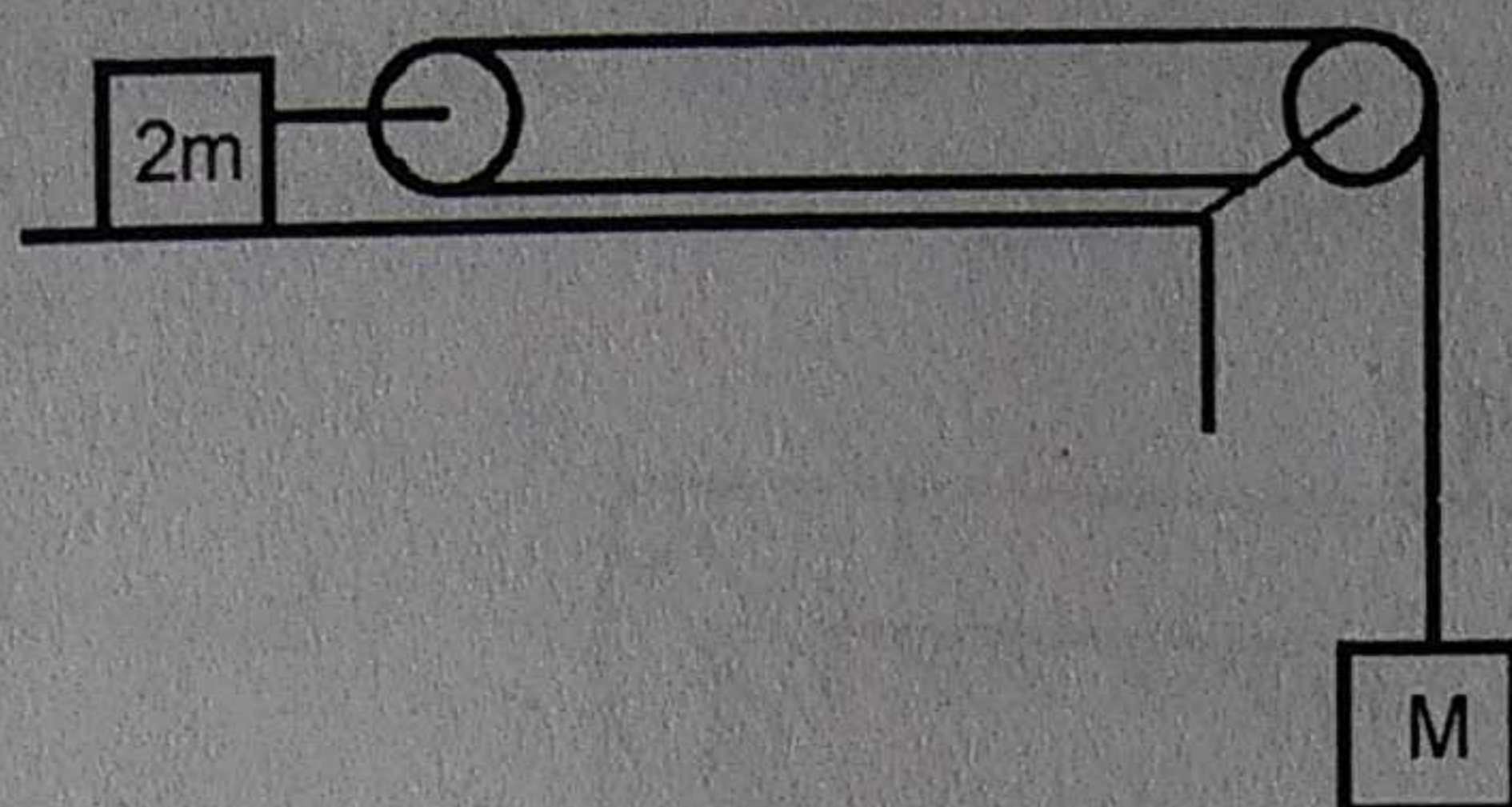


$$N = g \cos \theta$$

Q-6

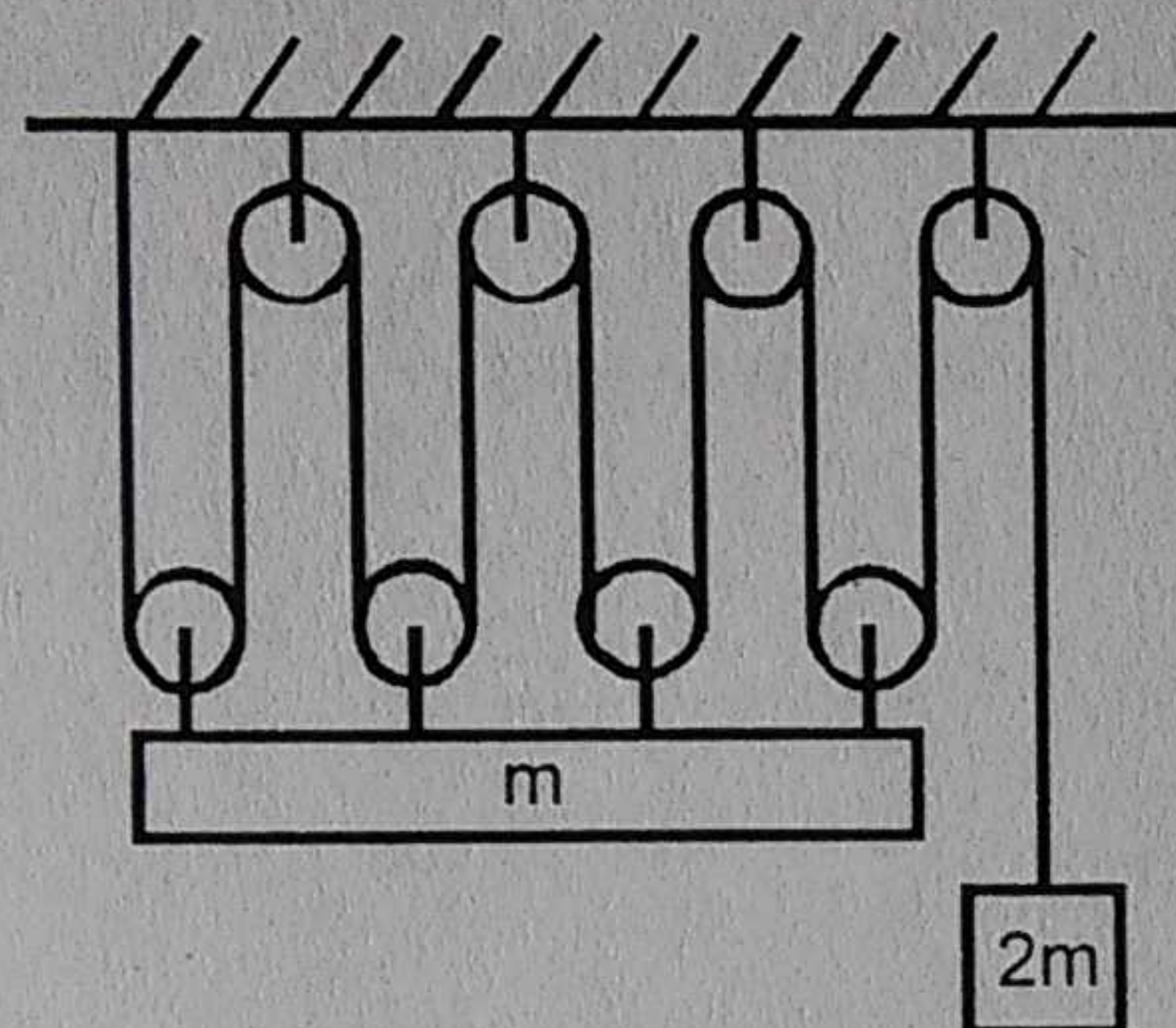


Q.-7

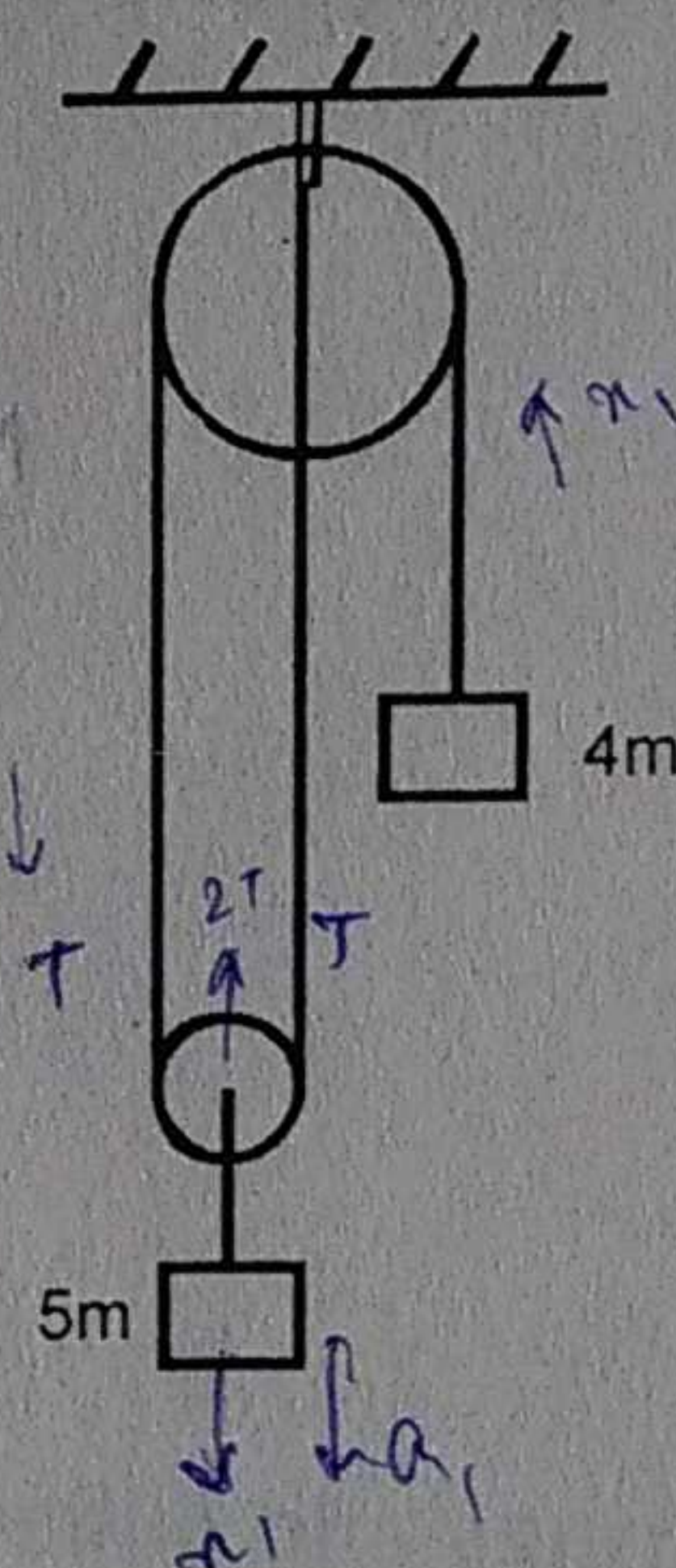


Q.-8

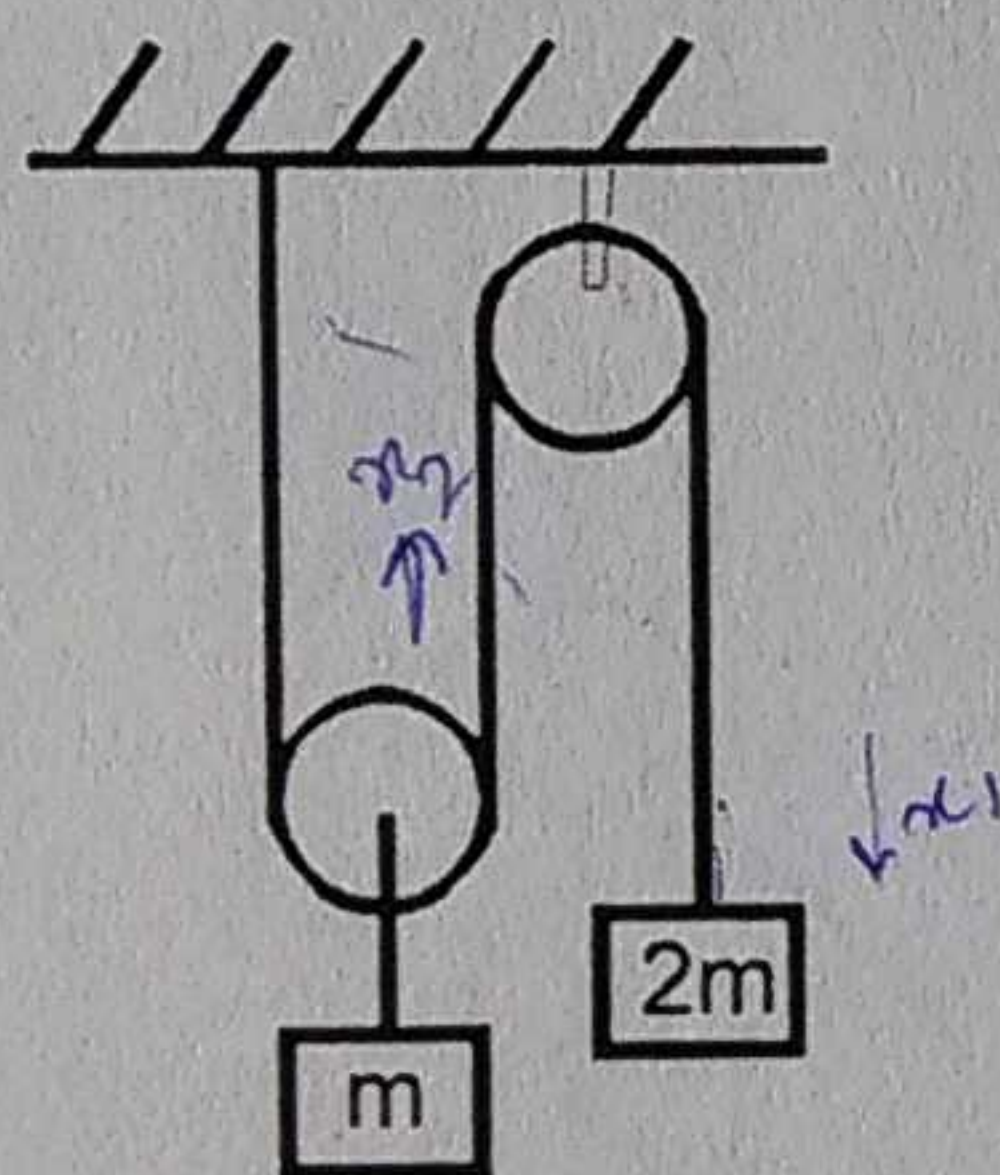
Q.-9



Q.-10



Q.-11



Q.12 Make a custom arrangement of blocks, pulleys and strings using your creativity. Assume some values of masses and solve the situation for acceleration of blocks.

Q.13 Make a custom arrangement of blocks, pulleys, strings and wedges using your creativity. Assume some values of masses of blocks and wedges and then solve the solution for acceleration of blocks and wedges.

1. $\frac{m_2 g}{2(m_1 + m_2)}$ rightwards

2. $\frac{g}{3}$ up the plane

3. $m \rightarrow \frac{g}{3}$ upwards

$2m \rightarrow \frac{2g}{3}$ Downwards

4. $5m \rightarrow \frac{g}{7}$ rightwards

$m \rightarrow \frac{g}{7}$ Downwards

$\frac{g}{7}$ rightwards

5. $m \rightarrow g \sin \theta$ (down the incline)

$\frac{g \sin^2 \theta \cdot \cos \theta}{3 + \sin^2 \theta}$ ($\perp$ to incline towards incline)

$3m \rightarrow \frac{g \sin \theta \cdot \cos \theta}{3 + \sin^2 \theta}$ rightwards

6. $m \rightarrow \frac{g}{3}$ upwards

$2m \rightarrow \frac{g}{3}$ downwards

7. $5m \rightarrow \frac{g \sin \theta \cdot \cos \theta}{(5 + \sin^2 \theta)}$

$m \rightarrow \frac{g}{3} \sin \theta$ (up the incline)

$\frac{g \sin^2 \theta \cos \theta}{(5 + \sin^2 \theta)}$ ($\perp$ to incline towards normal)

$2m \rightarrow \frac{g}{3} \sin \theta$ (down the incline)

$\frac{g \sin^2 \theta \cos \theta}{(5 + \sin^2 \theta)}$ ($\perp$ to incline Opp. to normal)

8. $M \rightarrow \frac{2g}{3}$ downwards

$2M \rightarrow \frac{g}{3}$ rightwards

9. Does your answer come out to be

$$m \rightarrow \frac{3g}{13} \text{ upwards}$$

$$2m \rightarrow \frac{24g}{13} \text{ downwards}$$

Is it possible?

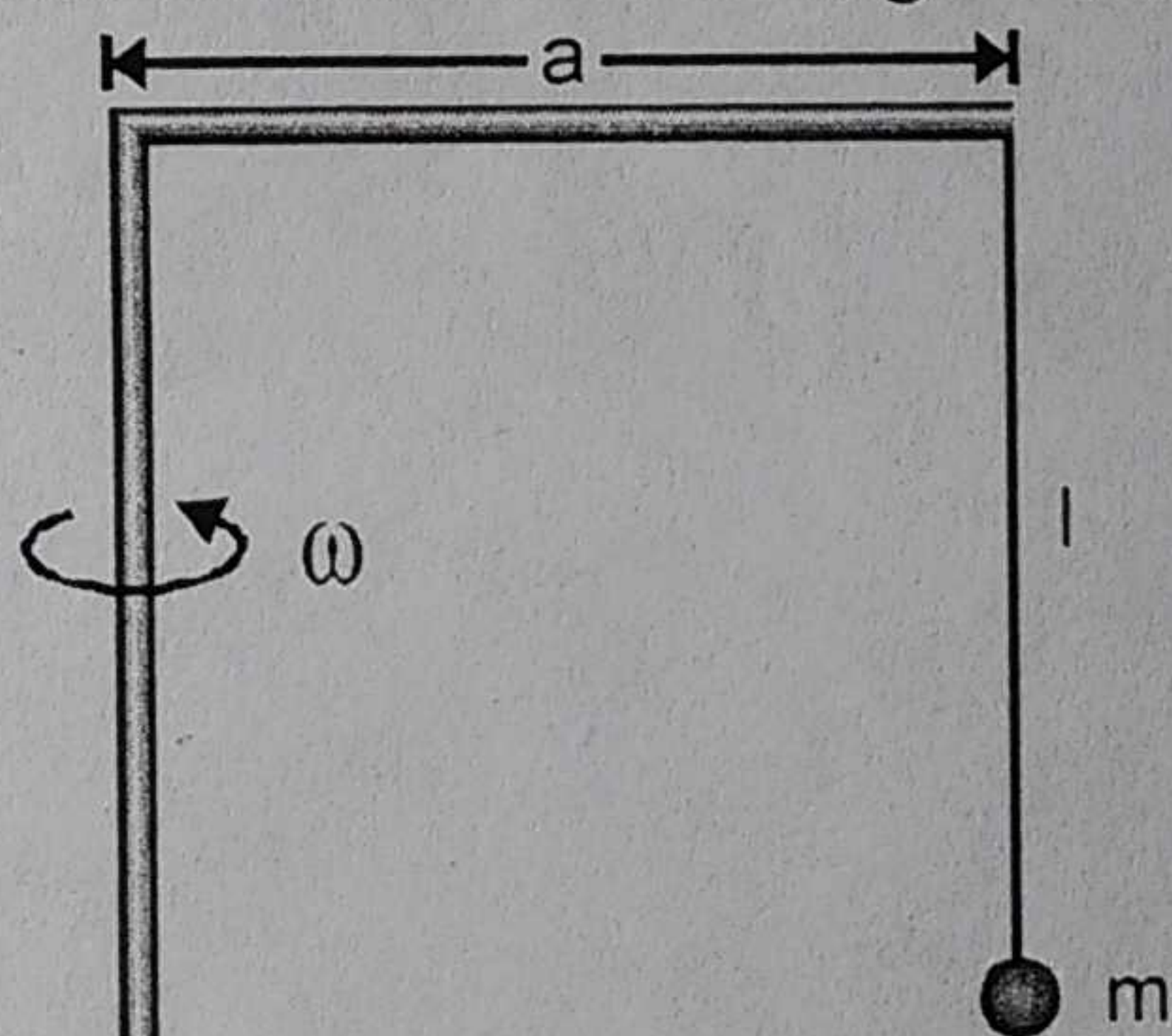
10. $4m \rightarrow \frac{2g}{7} \text{ downwards}$

$$5m \rightarrow \frac{g}{7} \text{ upwards}$$

11. $m \rightarrow \frac{g}{3} \text{ upwards}$

$$2m \rightarrow \frac{2g}{3} \text{ downwards}$$

1. A turn of radius 20 m is banked for the vehicle of mass 200 kg going at of speed of 10 m/s. Find the direction and magnitude of frictional force acting on a vehicle if it moves with a speed (a) 5 m/s (b) 15 m/s. Assume that friction is sufficient to prevent slipping $g=10 \text{ m/s}^2$.
2. A particle of mass m is placed over a horizontal circular table rotating with an angular velocity ' ω ' about a vertical axis passing through its centre. The distance of the object from the axis is r . Find the force of friction f between the particle and the table.
3. A simple pendulum is constructed by attaching a bob of mass m to a string of length L fixed at its upper end. The bob oscillates in a vertical circle. It is found that the speed of the bob is v when the string makes an angle α with the vertical. Find the tension in the string and the magnitude of net force on the bob at that instant.
4. A hemispherical bowl of radius R is rotating about its axis of symmetry which is kept vertical. A small block kept in the bowl rotates with the bowl without slipping on its surface. If the surface of the bowl is smooth and the angle made by the radius through the block with the vertical is α . Find the angular speed at which the bowl is rotating.
5. Is a body in uniform circular motion in equilibrium? *It has acc.*
6. (a) How many revolutions per minute must the apparatus shown in figure make about a vertical axis so that the cord makes an angle of 45° with the vertical?
(b) What is the tension in the cord then, given, $l = \sqrt{2} \text{ m}$, $a = 20 \text{ cm}$ and $m = 5.0 \text{ kg}$?



7. A car moves at a constant speed on straight but hilly road. One section has a crest and dip of the same 250 m radius. (a) As the car passes the crest the normal force on the car is one half the 16 kN weight of the car. What will be the normal force of the car as it passes through the bottom of the dip. (b) What is the greatest speed at which the car can move without leaving the road at the top of the hill? (c) Moving at a speed found in part (b) what will be the normal force on the car as it moves through the bottom of the dip?
8. A car driver going at speed v suddenly finds a wide wall at a distance r . Should he apply brakes or turn the car in a circle of radius r to avoid hitting the wall.
9. Show that the angle made by the string with the vertical in a conical pendulum is given by $\cos \theta = \frac{g}{L\omega^2}$, where L is the length of the string and ω is the angular speed.
10. A very small cube of mass m is placed on the inside of a funnel rotating about a vertical axis at a constant rate of n revolutions per sec. The wall of the funnel makes an angle θ with the horizontal. If the coefficient of static friction between the cube and funnel is μ and the centre of the cube is at a distance r from the axis of rotation, what are the largest and smallest values of n for which the block will not move with respect to the funnel?

11. A table with smooth horizontal surface is turning at an angular speed ω about its axis. A groove is made on the surface along a radius and a particle is gently placed inside the groove at a distance a from the centre. Find the speed of the particle with respect to the table as its distance from the centre becomes L .

LAWS OF MOTION PACKAGE PROBLEMS

- A. Page – 29 $\longrightarrow$ Illustration – 1
- B. Page – 30 $\longrightarrow$ Problem – 1
- C. Page – 30 $\longrightarrow$ Problem – 2
- D. Page – 31 $\longrightarrow$ Exercise – 4, Q-1
- E. Page – 31 $\longrightarrow$ Exercise – 4, Q-2
- F. Page – 31 $\longrightarrow$ Exercise – 4, Q-3
- G. Page – 31 $\longrightarrow$ Exercise – 4, Q-4
- H. Page – 31 $\longrightarrow$ Exercise – 4, Q-5
- I. Page – 31 $\longrightarrow$ Exercise – 4, Q-6
- J. Page – 31 $\longrightarrow$ Exercise – 4, Q-7

Answers

1.(a) $300\sqrt{5}N$ upward (b) $500\sqrt{5}N$ downwards

2. $mr\omega^2$

3. $T = m\left(g \cos \alpha + \frac{v^2}{L}\right), F_{net} = m\sqrt{g^2 \sin^2 \alpha + \frac{v^4}{L^2}}$

4. $\omega = \sqrt{\frac{g}{R \cos \alpha}}$

5. No

6. a) 27.6 b) 6.93 N

7. a) 24 kN b) 50 m/s c) 32 kN

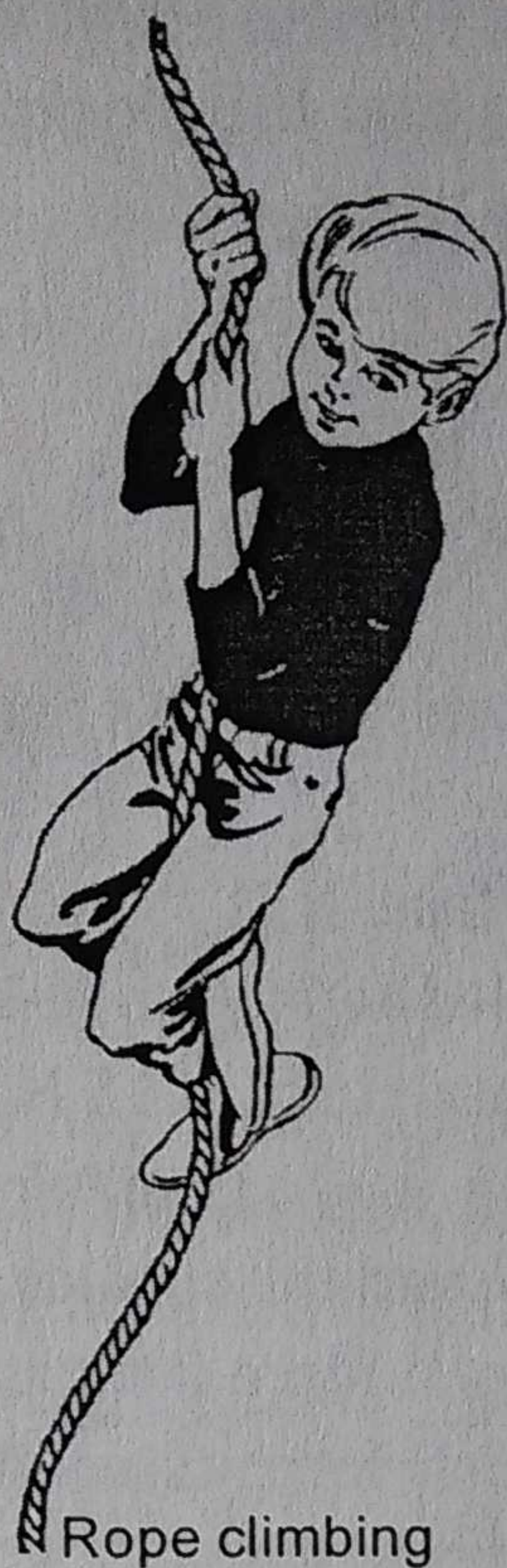
8. He should apply brakes

9. To Prove

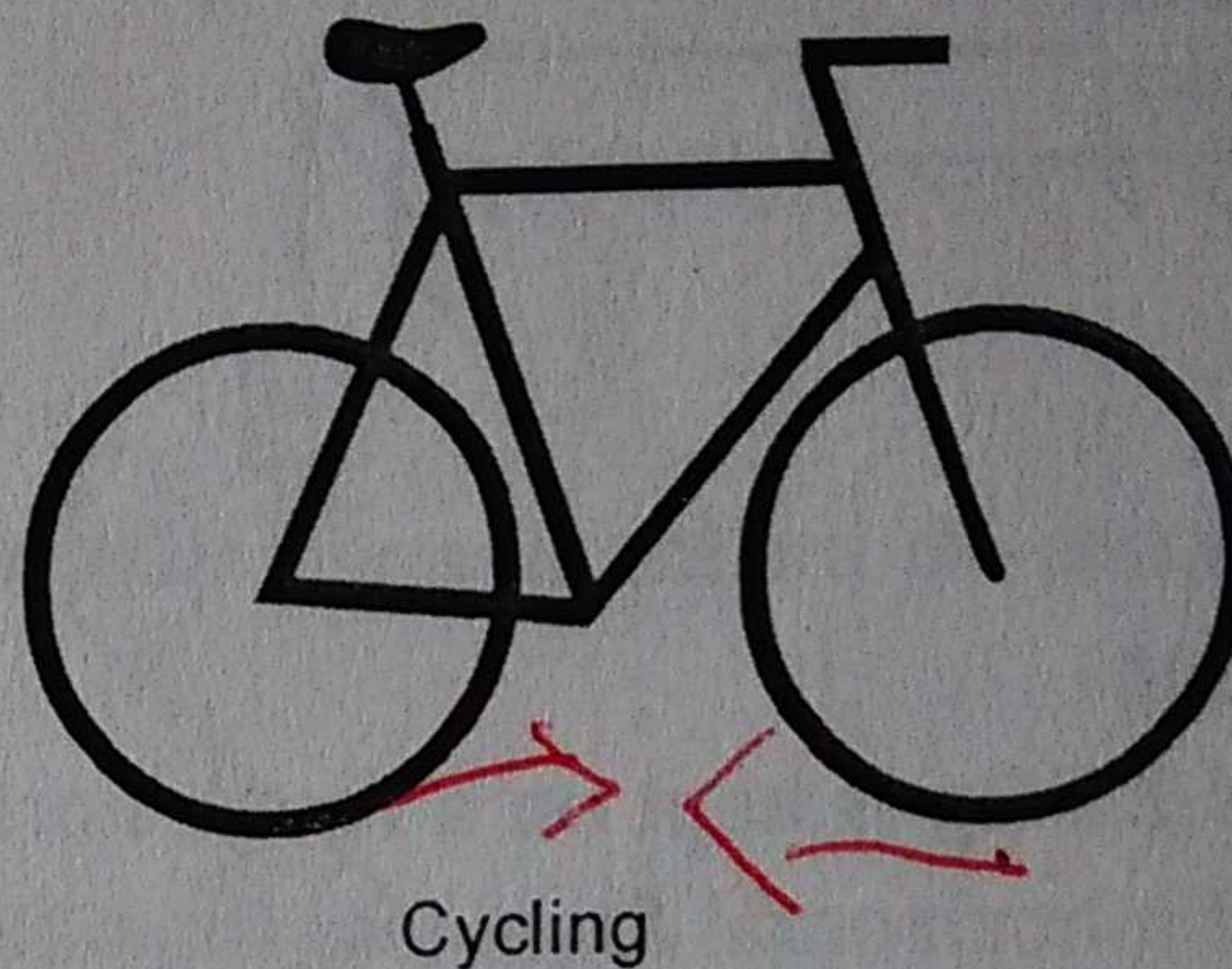
10. $n_{\max} = \frac{1}{2\pi} \sqrt{\frac{g(\sin \theta + \mu \cos \theta)}{r(\cos \theta - \mu \sin \theta)}}, n_{\min} = \frac{1}{2\pi} \sqrt{\frac{g(\sin \theta + \mu \cos \theta)}{r(\cos \theta - \mu \sin \theta)}}$

11. $v = \omega \sqrt{L^2 - a^2}$

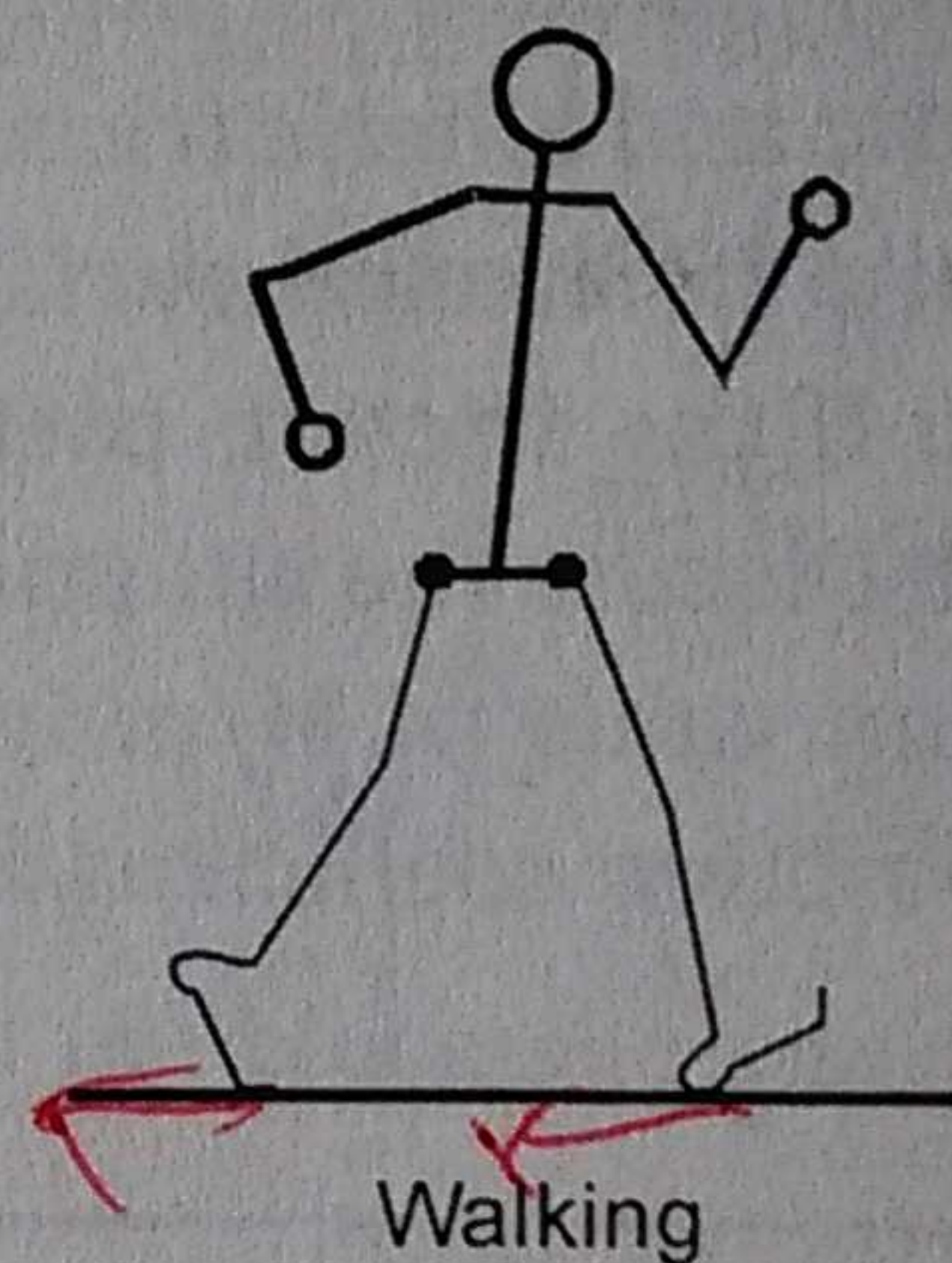
Analyze the role of friction (Draw FBD)



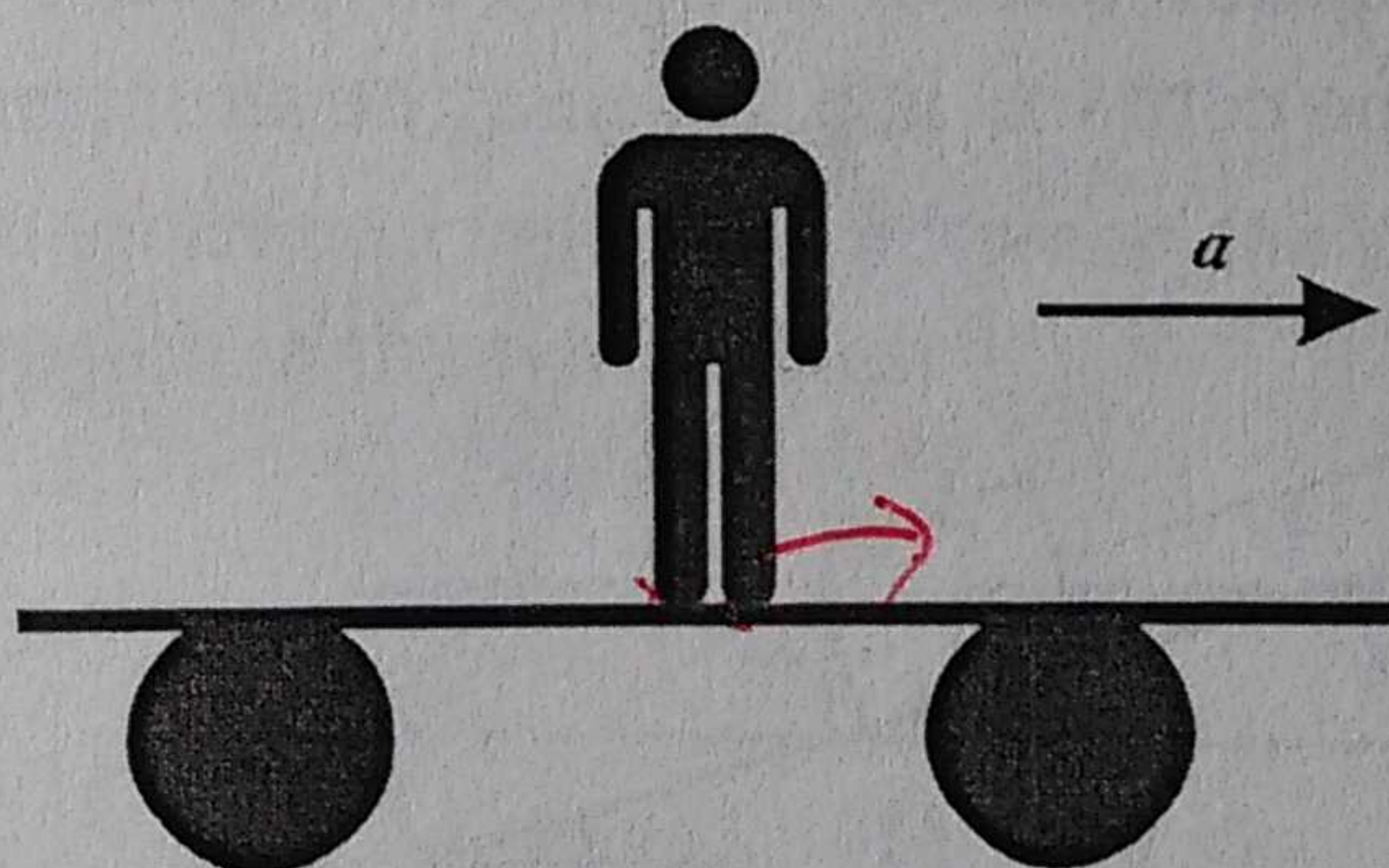
Rope climbing



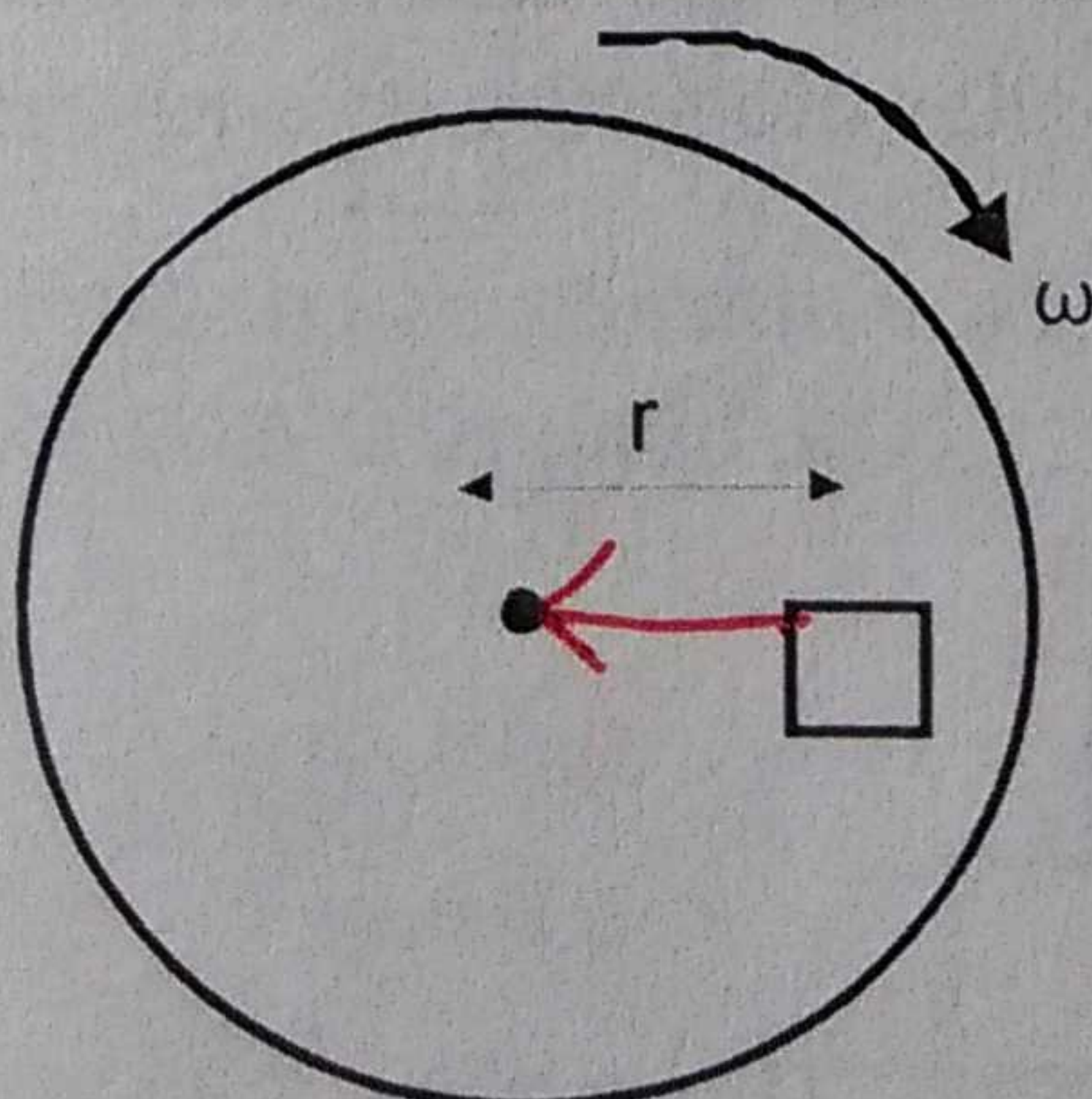
Cycling



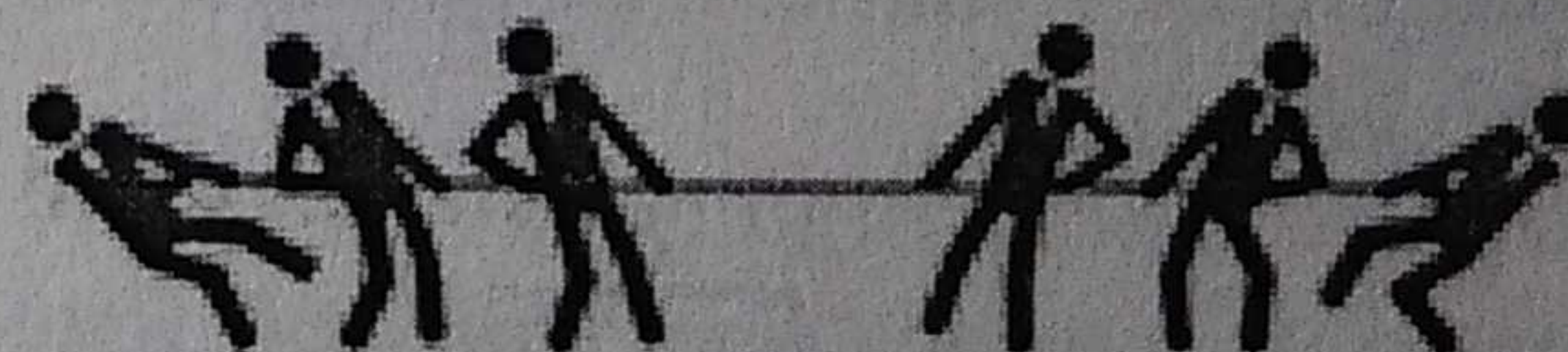
Walking



Man standing on accelerating trolley



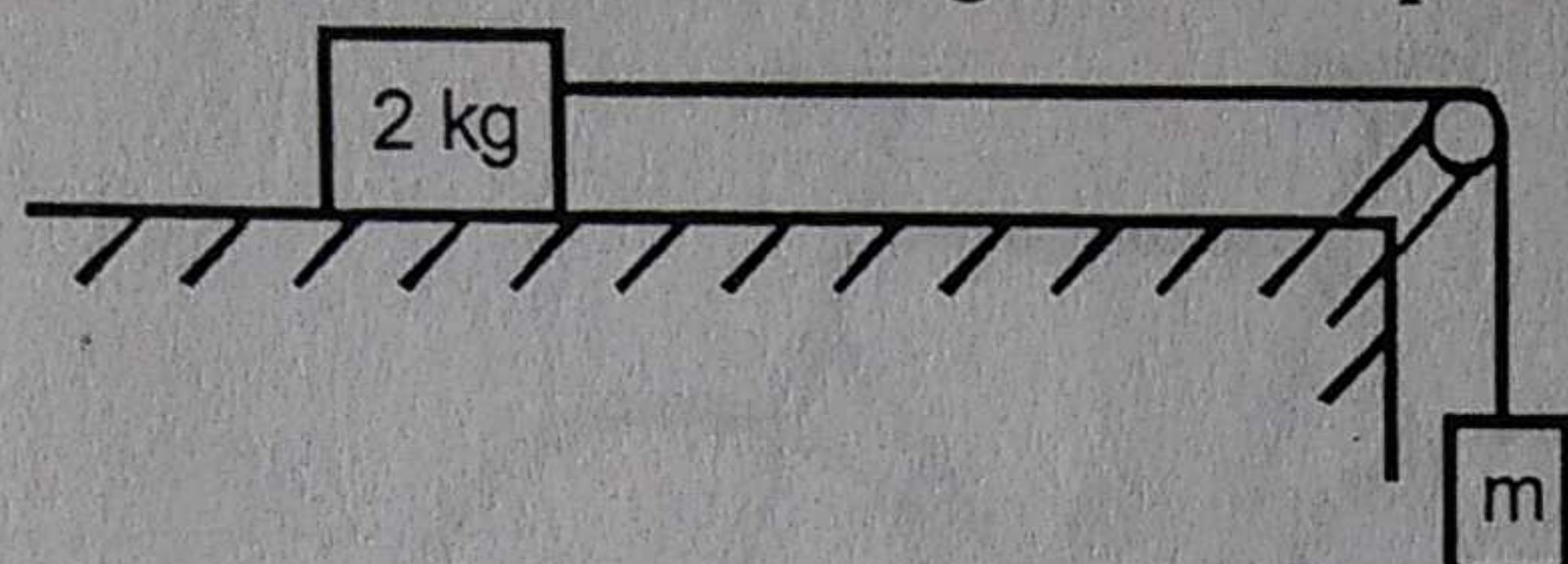
Block kept on rotating turn table



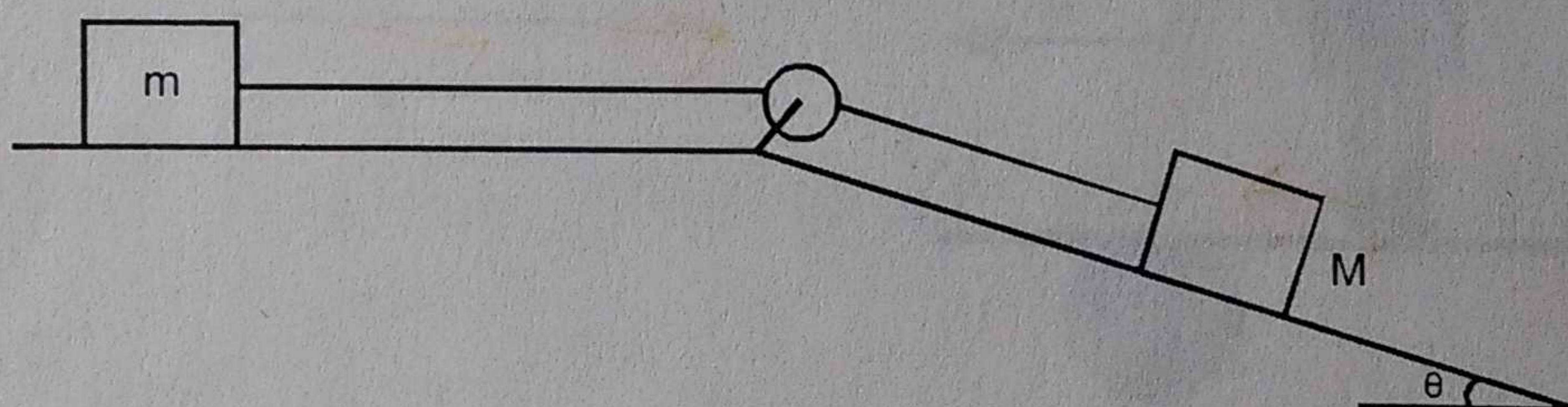
Tug of war

FRICTION

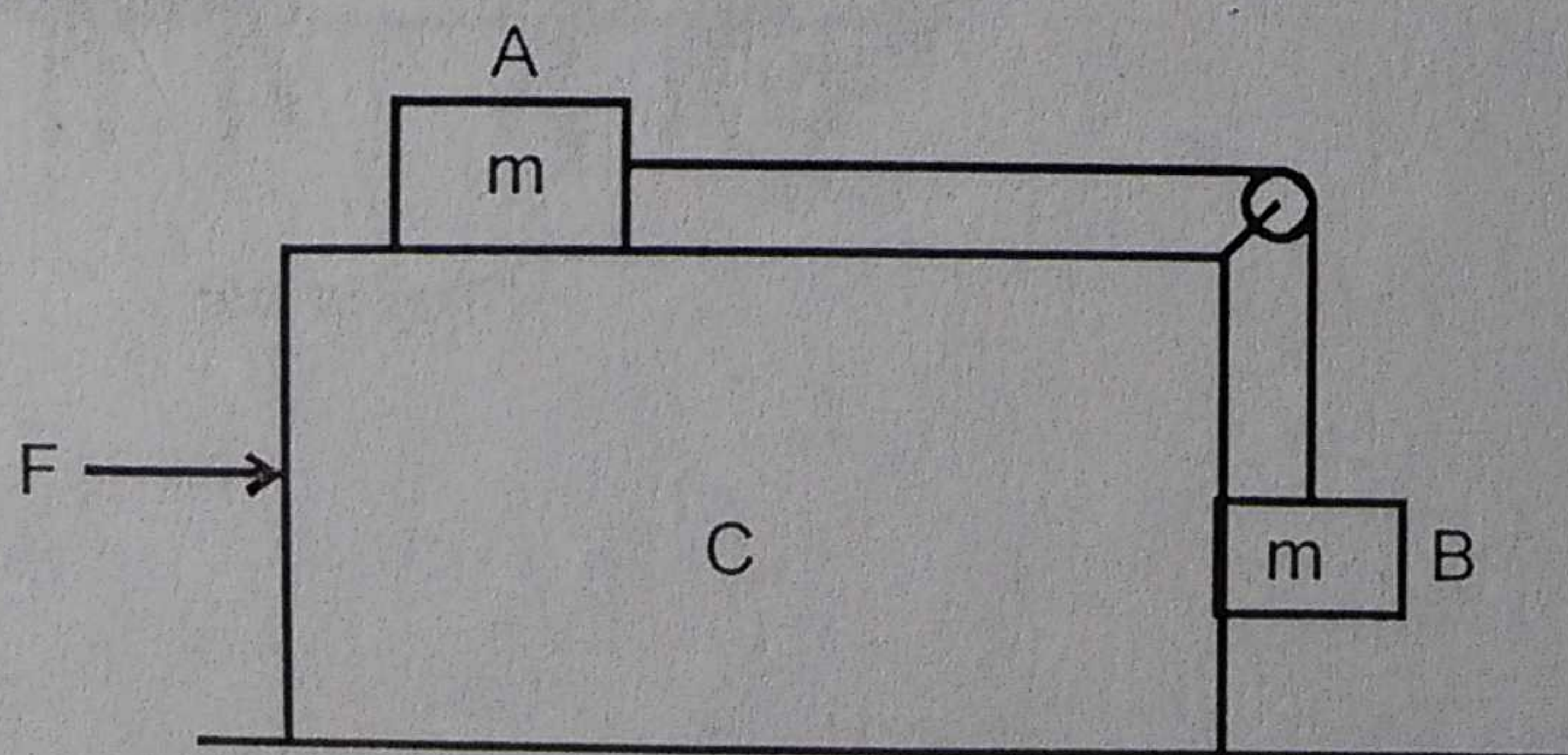
1. The coefficient of static friction between the block of 2 kg and the table shown in figure is $\mu_s = 0.2$. What should be the maximum value of m so that the blocks do not move? Take $g = 10 \text{ m/s}^2$. The string and the pulley are light and smooth.



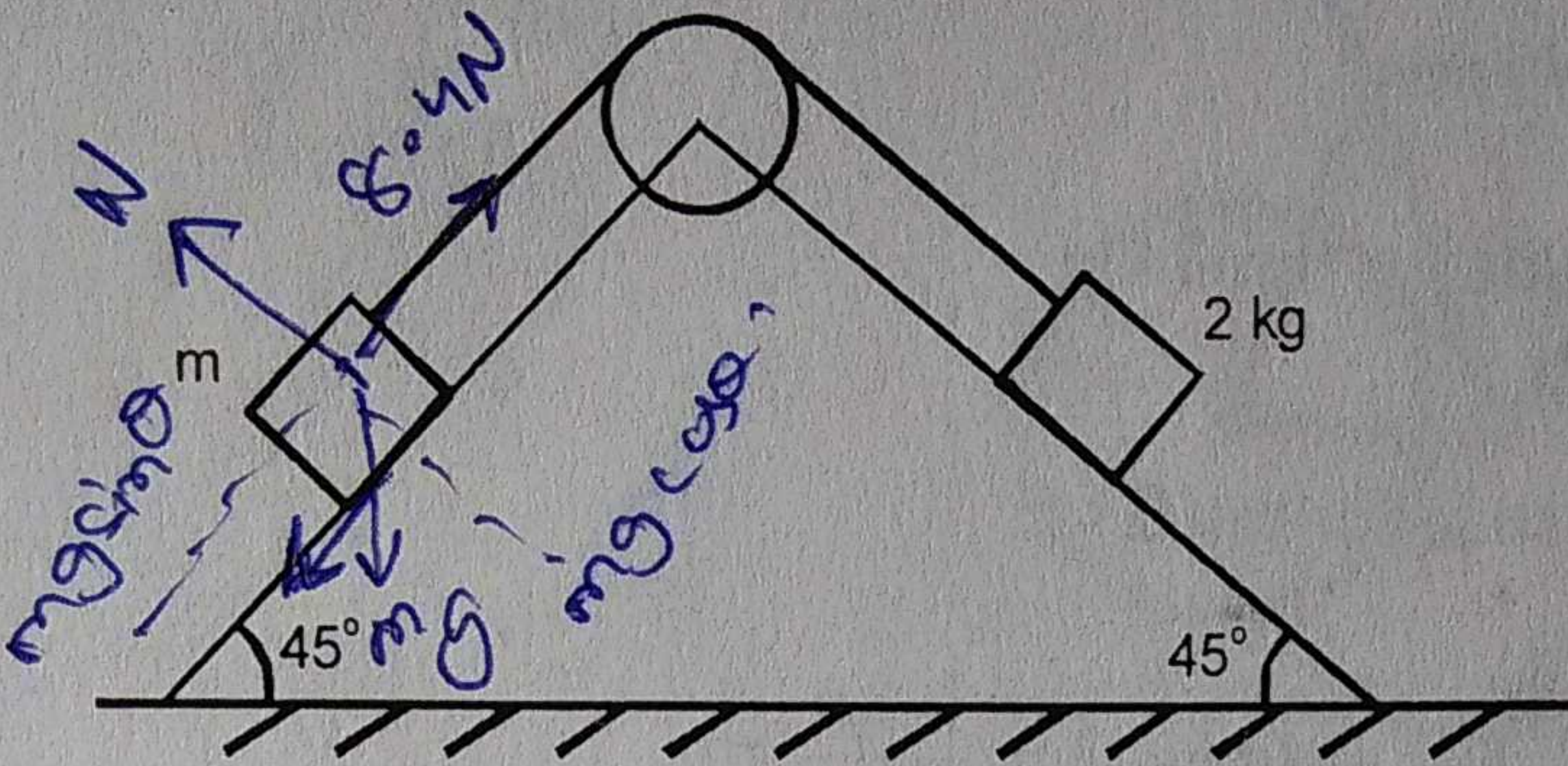
2. A block slides down on incline of angle 30° with an acceleration $g/4$. Find the kinetic friction coefficient.
3. A block of mass 2.5 kg is kept on a rough horizontal surface. It is found that the block does not slide if a horizontal force less than 15 N is applied to it. Also it is found that it takes 5 seconds to slide through the first 10 m if a horizontal force of 15 N is applied and the block is gently pushed to start the motion. Taking $g = 10 \text{ m/s}^2$, calculate the coefficients of static and kinetic friction between the block and the surface.
4. A block placed on a horizontal surface is being pushed by a force F making an angle θ with the vertical. If the friction coefficient is μ , how much force is needed to get the block just started. Discuss the situation when $\tan\theta < \mu$.
5. Find the maximum value of M/m in the situation shown in figure so that the system remains at rest. Friction coefficient at both the contacts is μ . Discuss the situation when $\tan\theta < \mu$



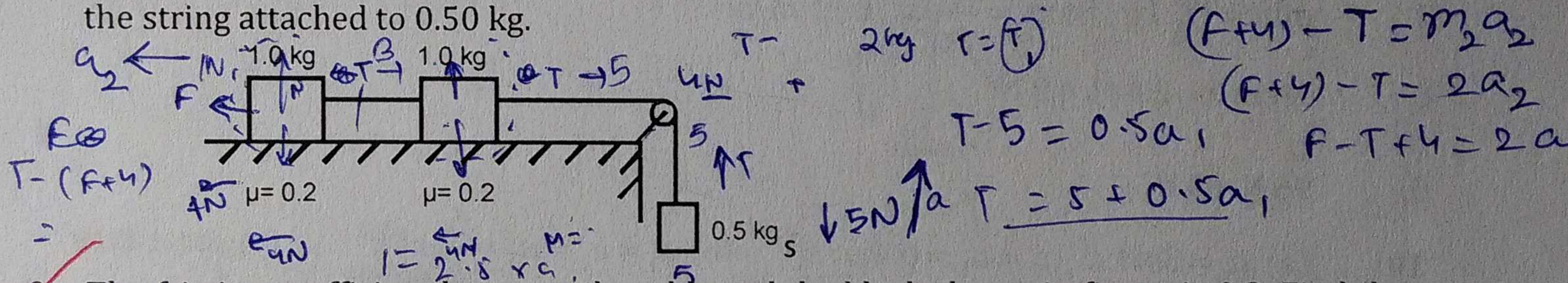
6. Consider the situation shown in figure. The horizontal surface below the bigger block is smooth. The coefficient of friction between the blocks is μ . Find the minimum and the maximum force F that can be applied in order to keep the smaller block at rest with respect to the bigger block.



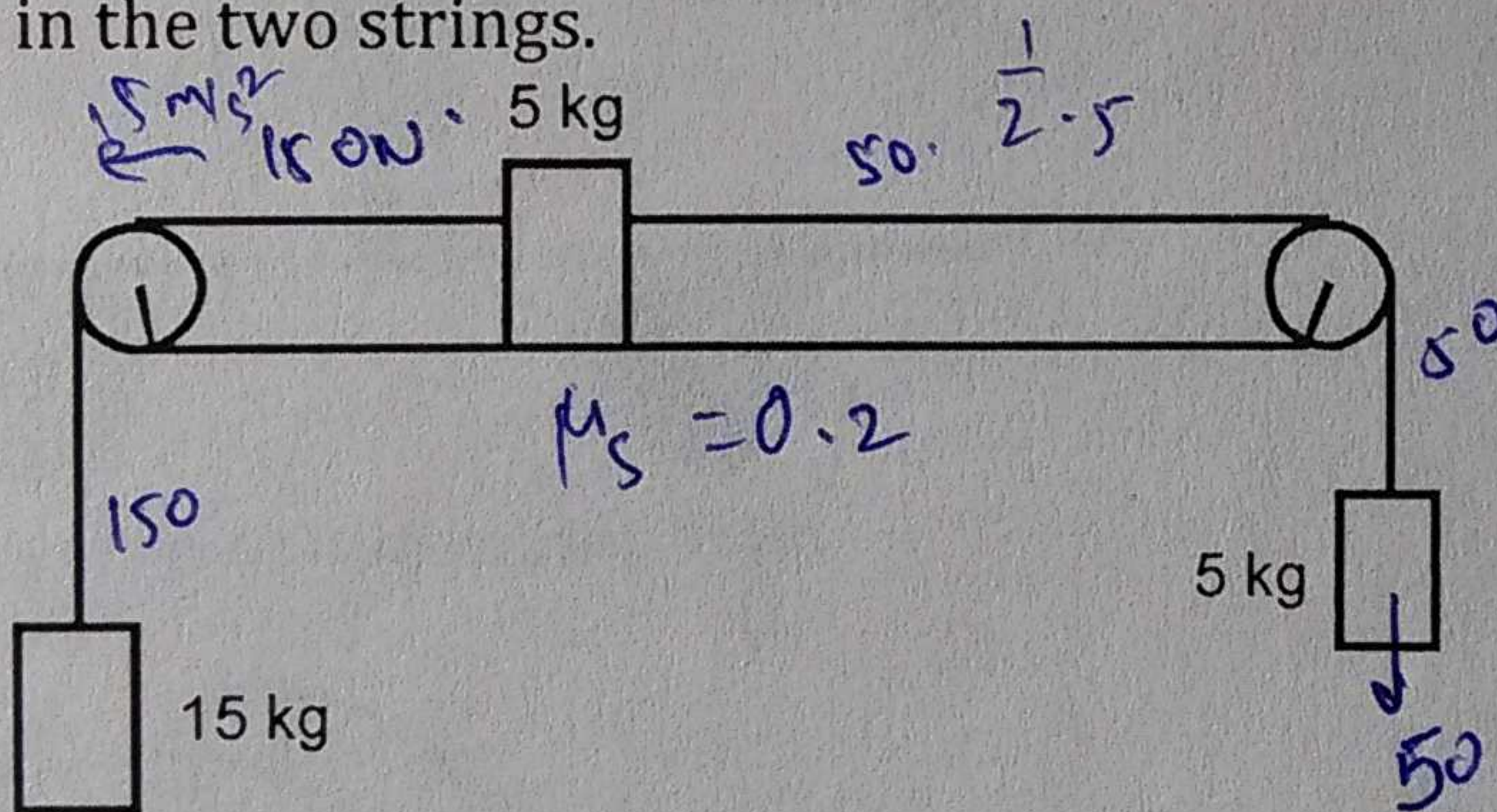
7. Figure shows two blocks connected by a light string placed on the two inclined parts of a triangular structure. The coefficients of static and kinetic friction are 0.28 and 0.25 respectively at each of the surface. (a) Find the minimum and maximum values of m for which the system remains at rest. (b) Find the acceleration of either block if m is given the minimum value calculated in the first part and is gently pushed up the incline for a short while.



8. Consider the situation shown in figure. Calculate (a) the acceleration of the 1.0 kg blocks, (b) the tension in the string connecting the 1.0 kg blocks and (c) the tension in the string attached to 0.50 kg.

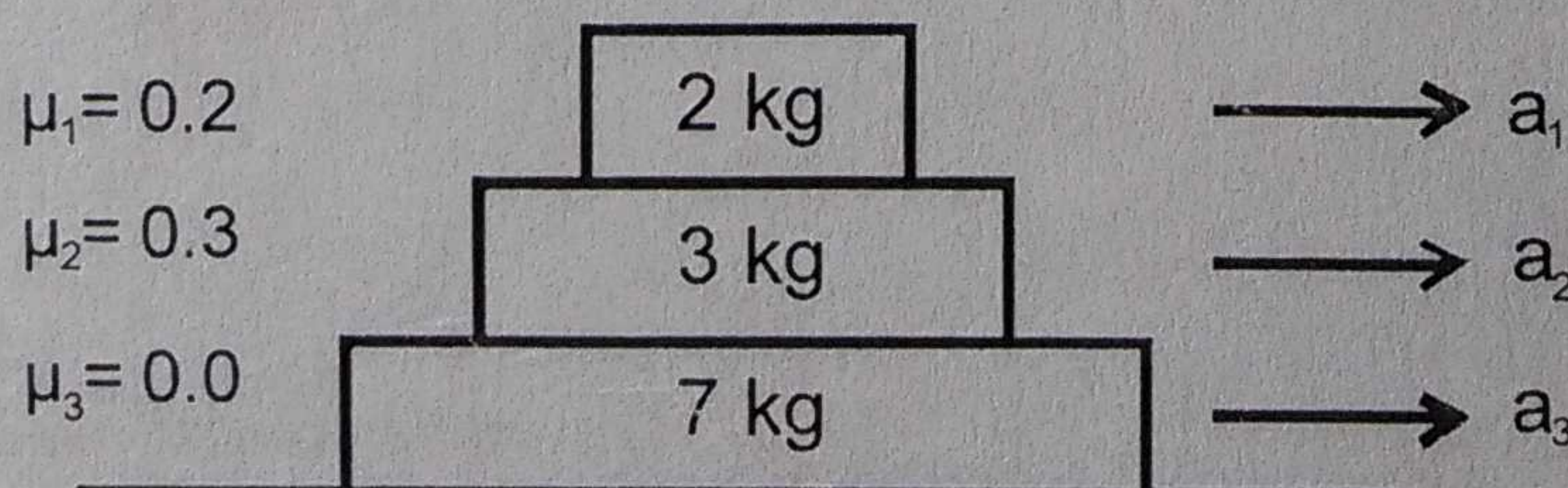


9. The friction coefficient between the table and the block shown in figure is 0.2. Find the tension in the two strings.



$$\begin{aligned} 100 &= 50 \\ 50 \times 0.2 &= 25 \text{ N} \\ 100 - 25 &= 75 \text{ N} \\ \frac{15}{2} &= m \cdot a \end{aligned}$$

10. Find the accelerations a_1 , a_2 , a_3 of the three blocks shown in figure if a horizontal force of 10 N is applied on (a) 2 kg block, (b) 3 kg block, (c) 7 kg block. Take $g = 10 \text{ m/s}^2$.



$$\frac{1}{2} = 0.5$$

$$0.04$$

ANSWERS

1. $m = 0.4 \text{ kg}$

2. $\frac{1}{2\sqrt{3}}$

3. $\frac{13}{25}$

4. $F = \frac{\mu mg}{\sin \theta - \mu \cos \theta}$, Will not slide for $\tan \theta < \mu$

5. $\frac{M}{m} = \frac{\mu}{\sin \theta - \mu \cos \theta}$, will not slide for $\tan \theta < \mu$

6. $F_{\min} = (M + 2m) \frac{1 - \mu}{1 + \mu} g$, $F_{\max} = (M + 2m) \frac{1 + \mu}{1 - \mu} g$

7. $\min = \frac{9}{8} \text{ kg}$, $\max = \frac{32}{9} \text{ kg}$, 0.31 m/s^2

8. 0.4 m/s^2 , 2.4 N , 4.8 N

9. 96 N , 68 N

10. (a) $a_1 = 3 \text{ m/s}^2$, $a_2 = a_3 = 0.4 \text{ m/s}^2$ (b) $a_1 = a_2 = a_3 = \frac{5}{6} \text{ m/s}^2$ (c) same as (b)